

Chapter 35 Interference

Thomas Young (1773 –1829) was a British polymath and physician. Young made notable scientific contributions to the fields of vision, light, solid mechanics, energy, physiology, language, musical harmony, and Egyptology.

1. Young's modulus (杨氏模量)
2. The founder of physiological optics
3. The first to describe astigmatism (散光)
4. important contribution to haemodynamics (血液动力学)
5. compared the grammar and vocabulary of 400 languages
6. decipherment of Egyptian hieroglyphs (specifically the Rosetta Stone)
7. Young temperament (tuning musical instruments)
8. In Young's own judgment, **the most important achievement was to establish the wave theory of light.**



"The Last Man Who Knew Everything".

Chapter 35 Interference



Optical Path and Optical Path Difference
光程和光程差

Today's topics (Chapter 35)

- Interference of Waves
- Double-Slit Experiment
- Thin-Film Interference
- Michelson's Interferometer

First war between wave and particle



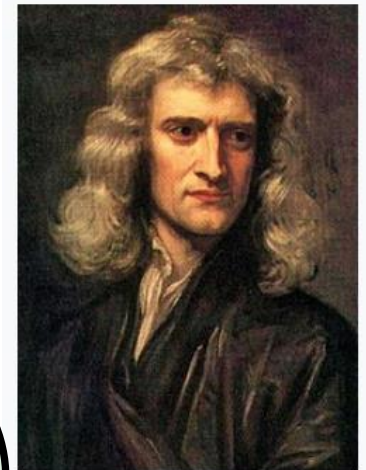
Robert Hooke (1635 – 1703): In *micrographia* 《显微术》

Light is wave!

Isaac Newton (1642 – 1726): in 1672, Newton found white light can be decomposed into seven colors. This dispersion is the mixture and division of colored particles.

The mixture of colors is stolen from my work in 1665. Your particle theory is completely s**t.

Light is particle!

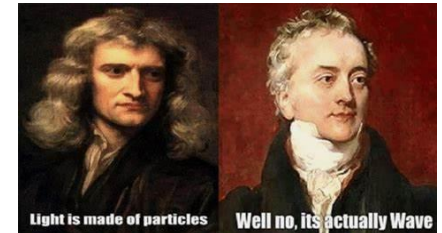
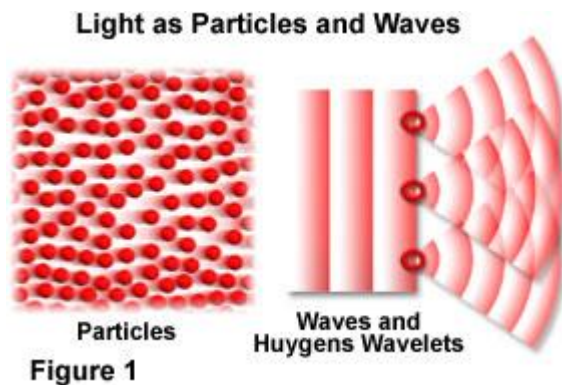


I was just standing on the shoulders of giants

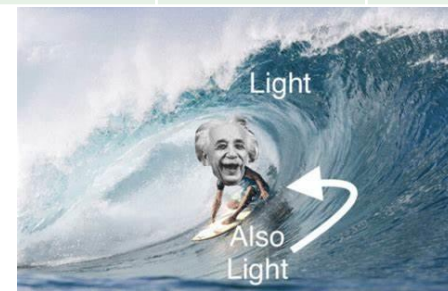


Particle-wave duality of light

LIGHT IS A
Wave!

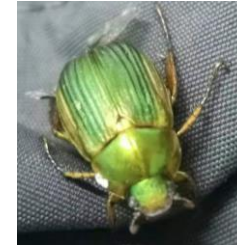
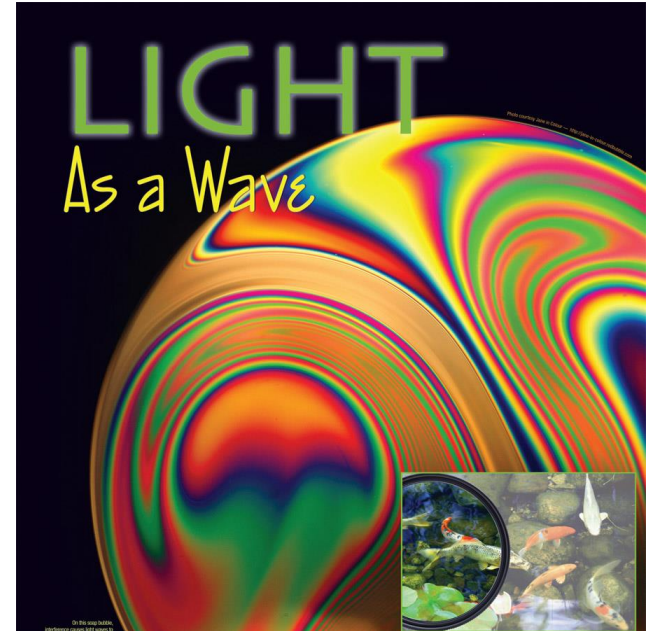
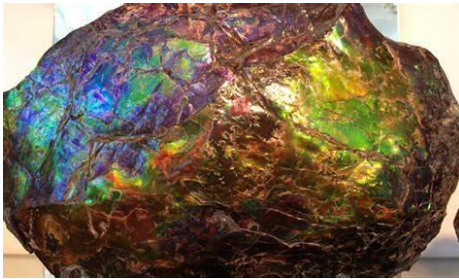
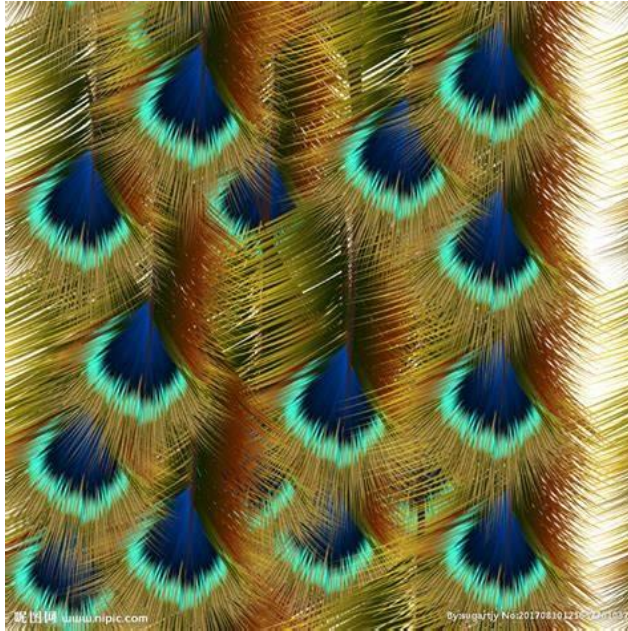


Reflection	😊	😊
Refraction	😊	😊
Interference	X	😊
Diffraction	X	😊
Polarization	X	😊
Photoelectric effect	😊	X



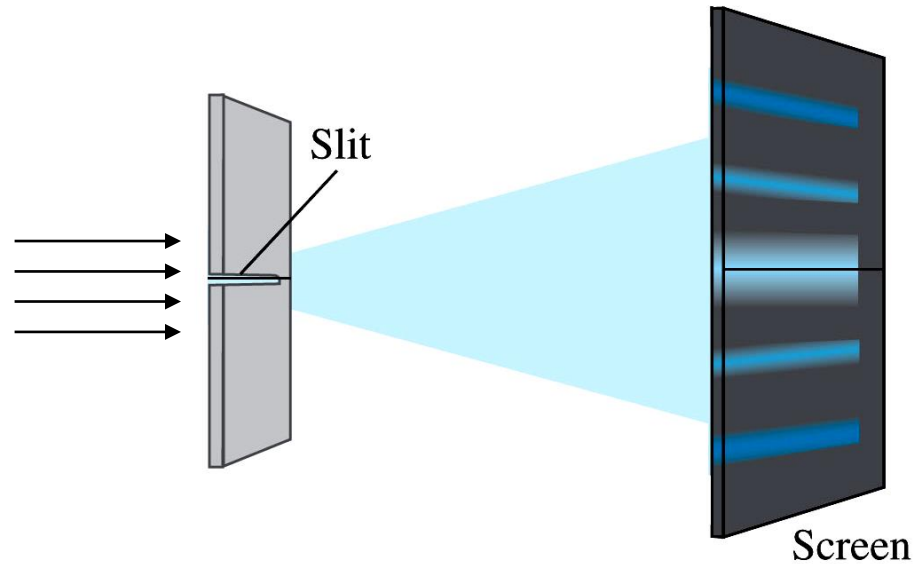
Einstein illustrates how light is both a wave and a particle (1905)

Light as a wave



We can see color in bird feathers, insects and soap bubbles because light is a wave – its interference and diffraction gives rise to the colorful world

Geometric and physical optics

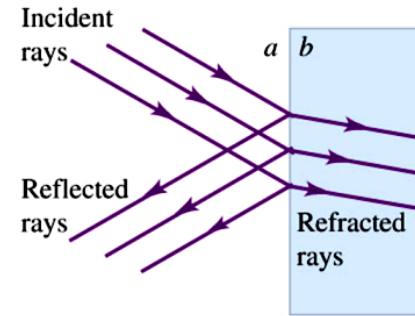


If objects' size is much larger than the wavelength of light, one can treat light as rays that travel in straight lines and obey the simple reflection and refraction law. This realm is called geometric optics where the wave nature of light can be neglected. However, as the objects' size becomes comparable to the wavelength of light, one needs to take into account the wave nature of light, and phenomena related to such nature, such as diffraction and interference, becomes prominent. This realm is called physical optics.

Interference and diffraction

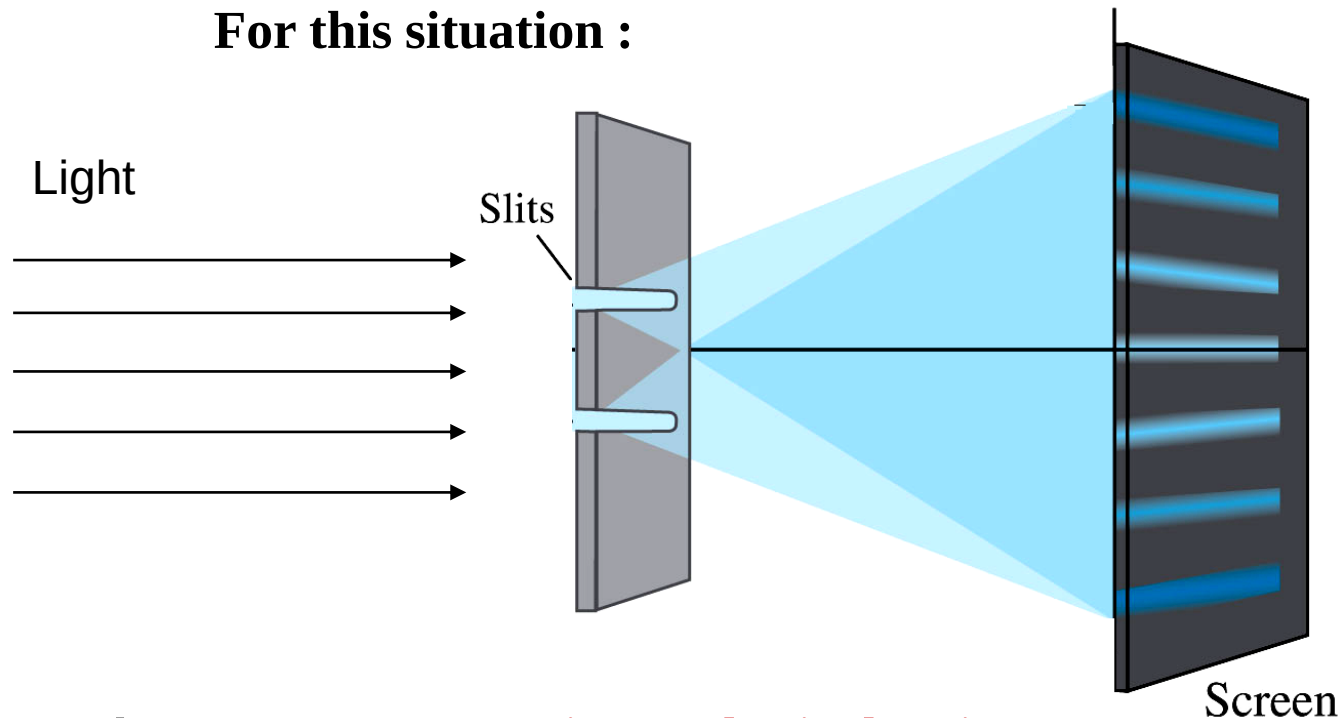
Geometric optics, ray optics :

We represent light as **rays**, straight lines that are bent at a reflecting or refracting surface



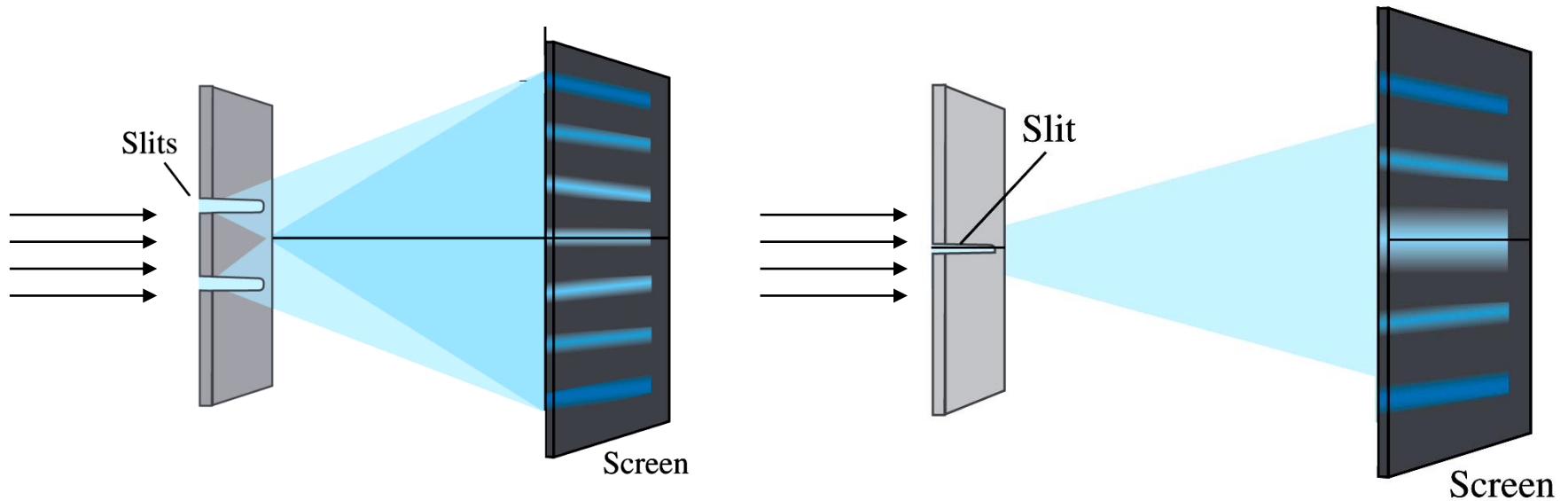
ok!

For this situation :



We have to use **wave optics or physical optics**

Interference and diffraction

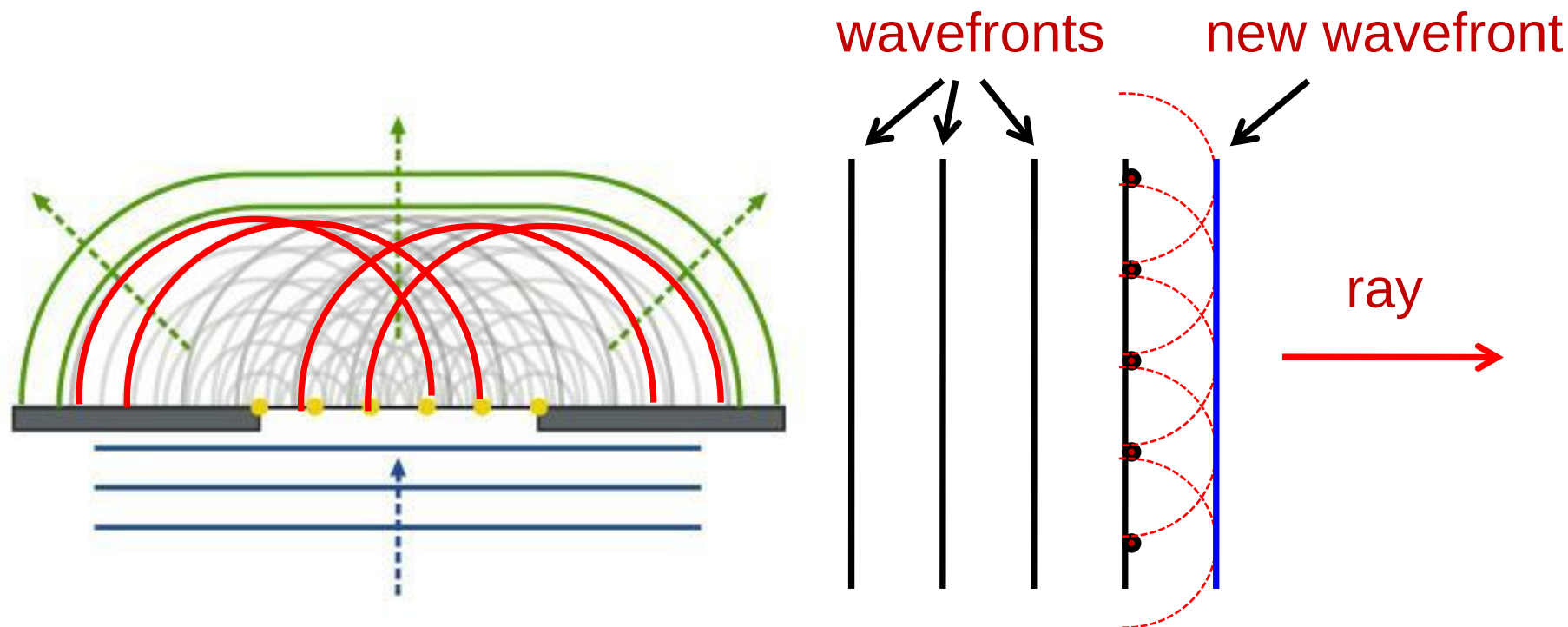


Interference occurs when waves from different sources encounter each other in space.

Diffraction occurs when propagation of a wave is disturbed by objects whose size is comparable to the wavelength.

Both **interference** and **diffraction** are consequences of superposition and Huygen's principle

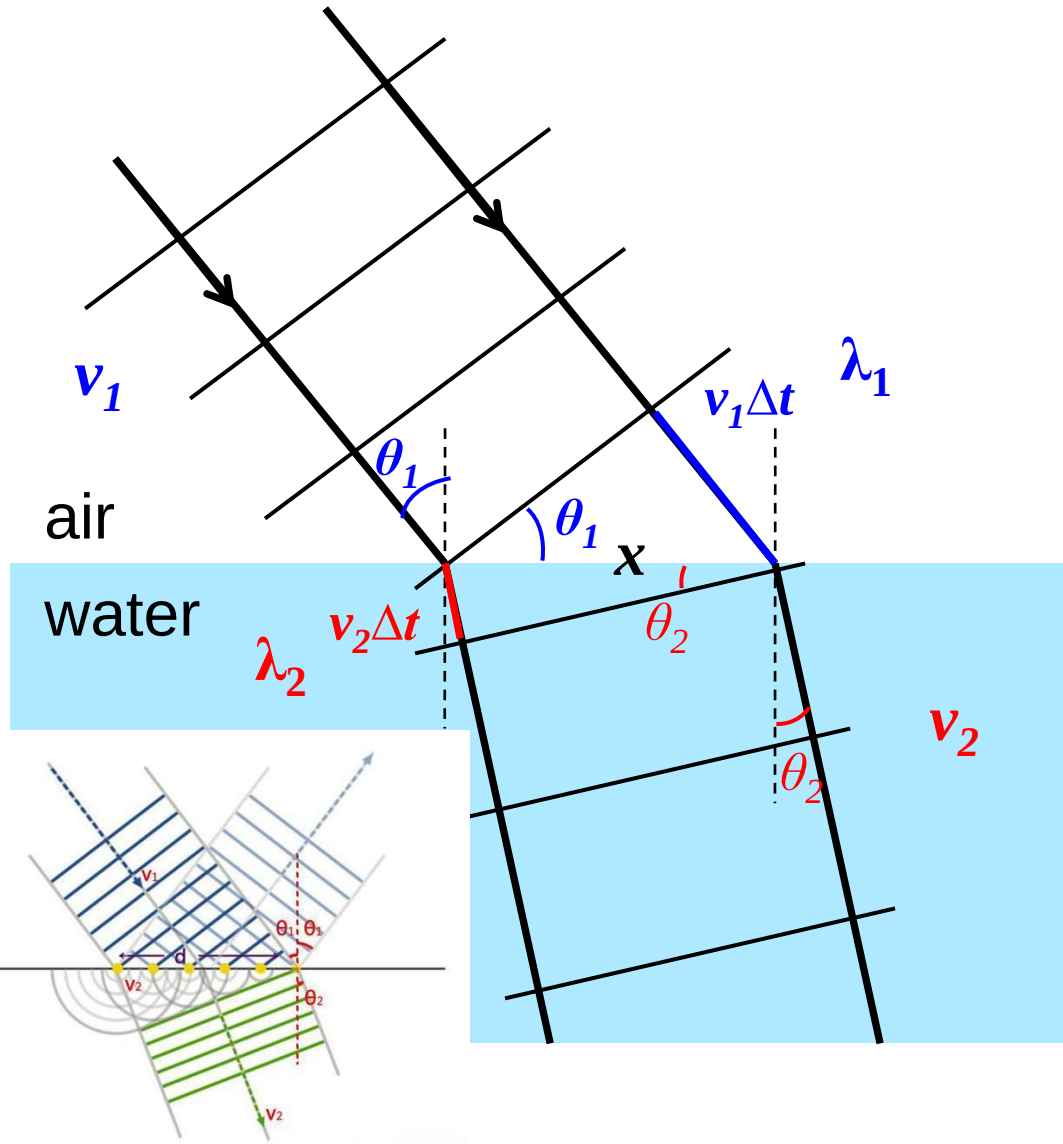
Huygen's principle



Huygen's principle : Every point of a wavefront may be considered **the point source of secondary wavelets** that spread out in all directions with a speed equal to the speed of propagation of the wave. The new position of the wavefront will be that of a surface tangent to these secondary wavelets.

Huygen's principle is just a convenient way to visualize the light propagation. **Maxwell's equation** are the fundamental relationships for EM wave propagation.

Huygen's principle and law of refraction



$$\sin \theta_1 = \frac{v_1 \Delta t}{x}$$

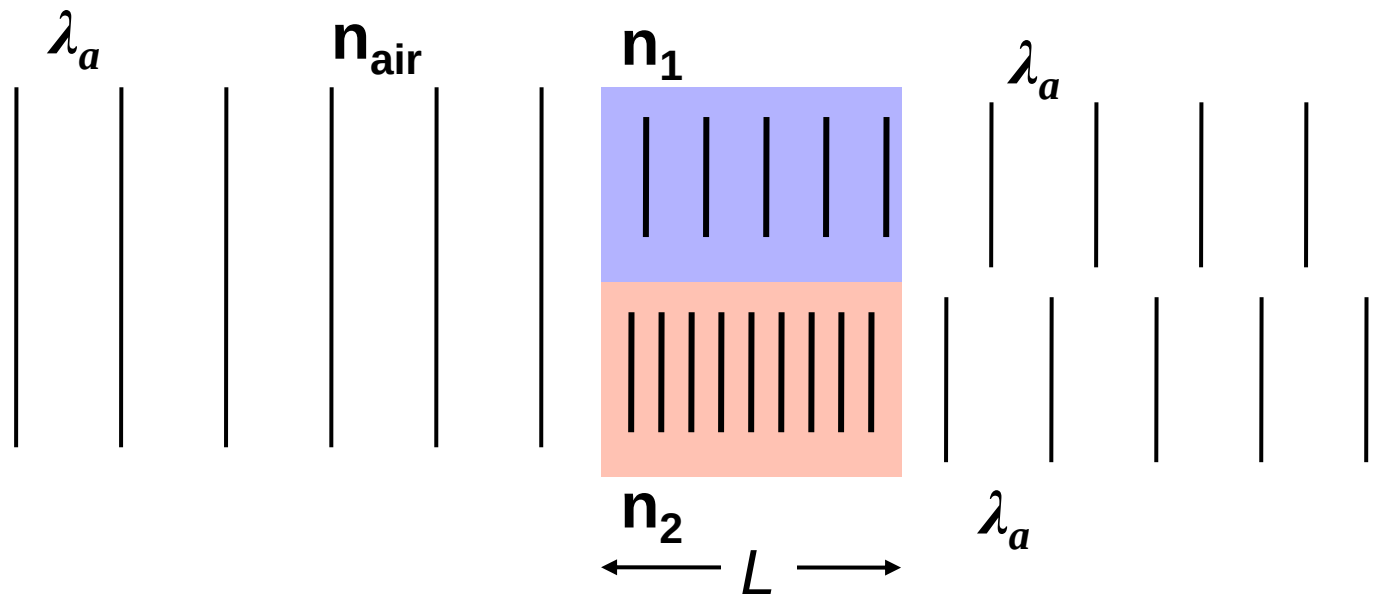
$$\sin \theta_2 = \frac{v_2 \Delta t}{x}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

$$= \frac{c/n_1}{c/n_2}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Phase difference and path length difference



No. of wavelength N_1 in medium 1 : $\frac{L}{\lambda_{n1}} = \frac{Ln_1}{\lambda_a}$

No. of wavelength N_2 in medium 2 : $\frac{L}{\lambda_{n2}} = \frac{Ln_2}{\lambda_a}$

Phase difference between the waves: $\frac{Ln_2}{\lambda_a} - \frac{Ln_1}{\lambda_a} = \frac{L}{\lambda_a} (n_2 - n_1)$

say, if it is 3.6
we take the
decimal fraction,
i.e. 0.6

Phase difference and path length difference

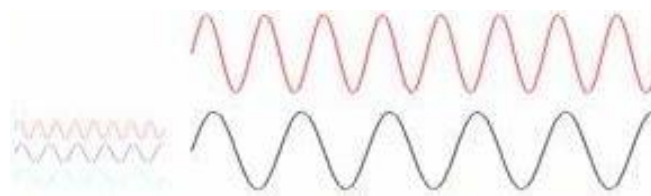
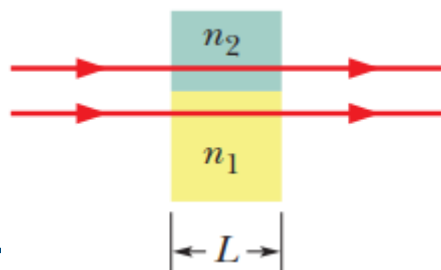
The *optical path length* (**OPL**) is the product of the geometric length of the path and the index of refraction. The *optical path difference* (**OPD**) is then

$$\text{OPL} = \sum L_i n_i; \quad \text{OPD} = L_2 n_2 - L_1 n_1$$

How OPD compares with λ determines whether the interference is constructive or destructive:

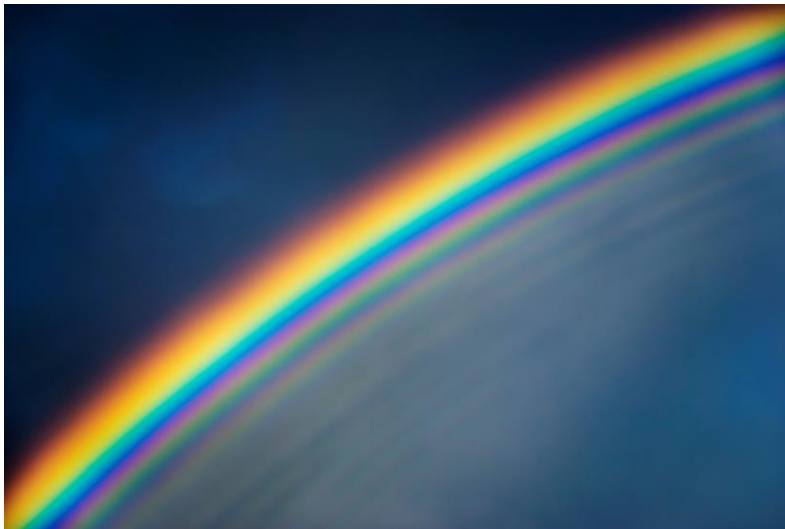
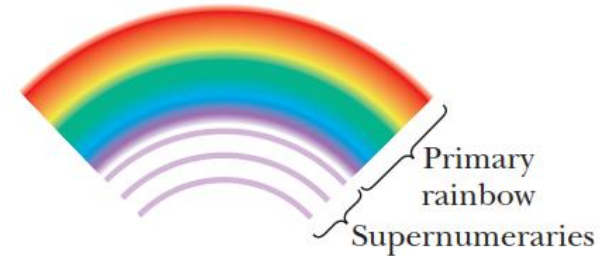
$$\frac{\Delta L}{\lambda} = 0, 1, 2, \dots \quad (\text{fully constructive interference})$$

$$\frac{\Delta L}{\lambda} = 0.5, 1.5, 2.5, \dots \quad (\text{fully destructive interference})$$



Rainbows revisited: supernumerary arcs

If you are lucky and look carefully below a primary rainbow, you can see dimmer colored arcs called supernumeraries. Like the main arcs of the rainbow, the supernumeraries are due to waves that emerge from each drop approximately in phase with one another to give constructive interference. The number of visible supernumeraries depends upon the size of the raindrops



Two sources S_1 and S_2 oscillating in phase emit sinusoidal waves. Point P is 7.3 wavelengths from source S_1 and 4.3 wavelengths from source S_2 . As a result, at point P there is



- A. constructive interference.
- B. destructive interference.
- C. neither constructive nor destructive interference.
- D. not enough information given to decide

Two sources S_1 and S_2 oscillating in phase emit sinusoidal waves. Point P is 7.3 wavelengths from source S_1 and 4.6 wavelengths from source S_2 . As a result, at point P there is

A. constructive interference.

B. destructive interference.



C. neither constructive nor destructive interference.

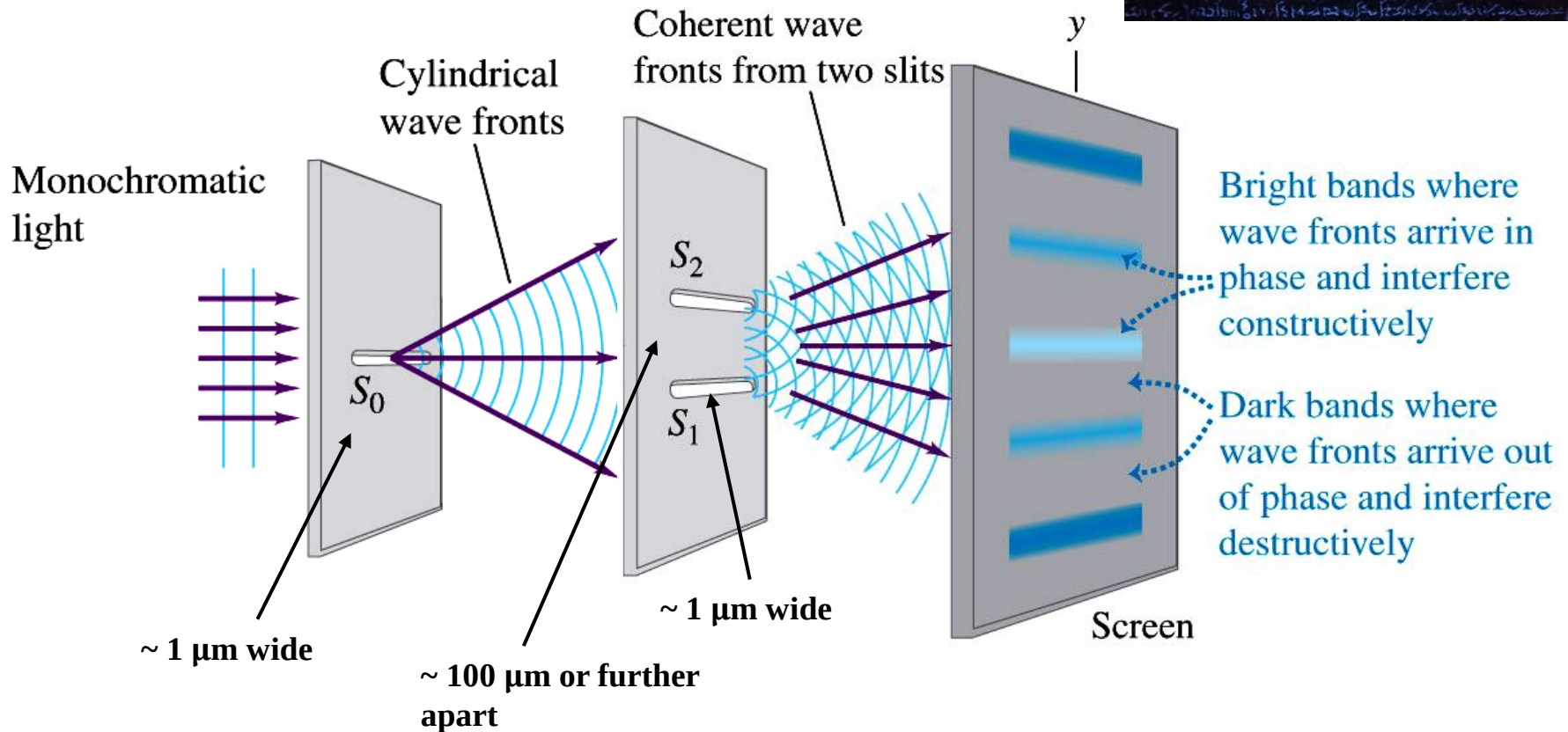
D. not enough information given to decide

Today's topics (Chapter 35)

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- Thin-Film Interference
- Michelson's Interferometer

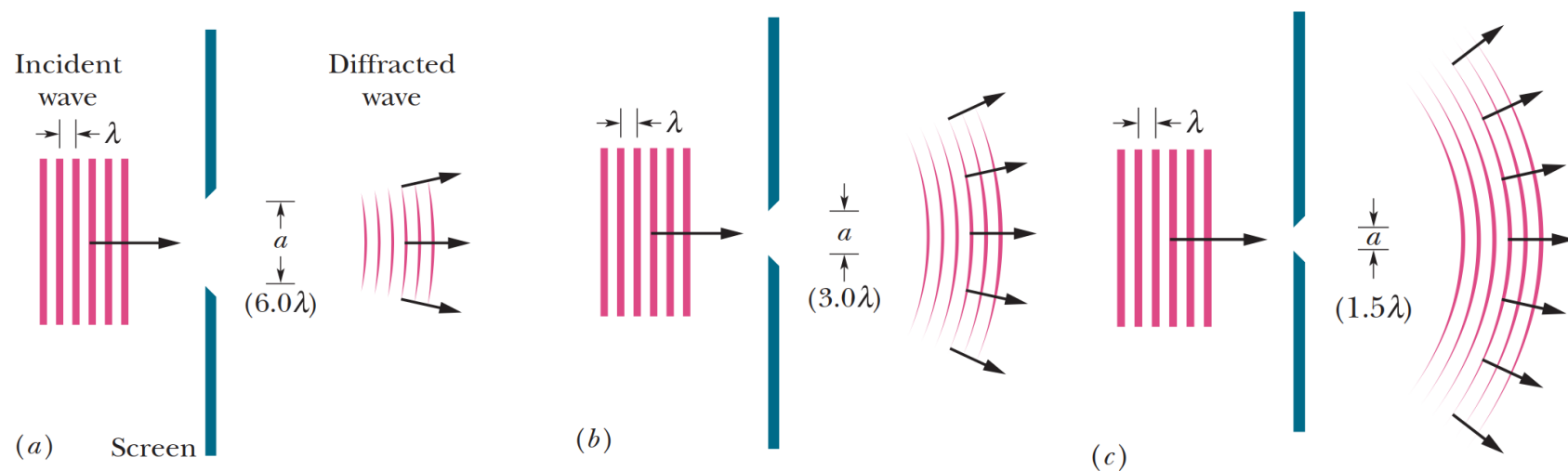
Young's experiment

Interference of light waves passing through two slits



Young's Interference experiment

In 1801, Thomas Young experimentally proved that light is a wave, contrary to what most other scientists then thought. He did so by demonstrating that light undergoes interference, as do water waves, sound waves, and waves of all other types. In addition, he was able to measure the average wavelength of sunlight. Huygen's principle yields that light will diffract through the slit, creating circular wavefronts.



Young's Interference experiment

After diffracted by the first slit, light reaches screen B, it is diffracted by slits S_1 and S_2 , which then act as two point sources of light. The light waves traveling from slits S_1 and S_2 overlap and undergo interference, forming an interference pattern of maxima and minima on viewing screen C.

Some terms:

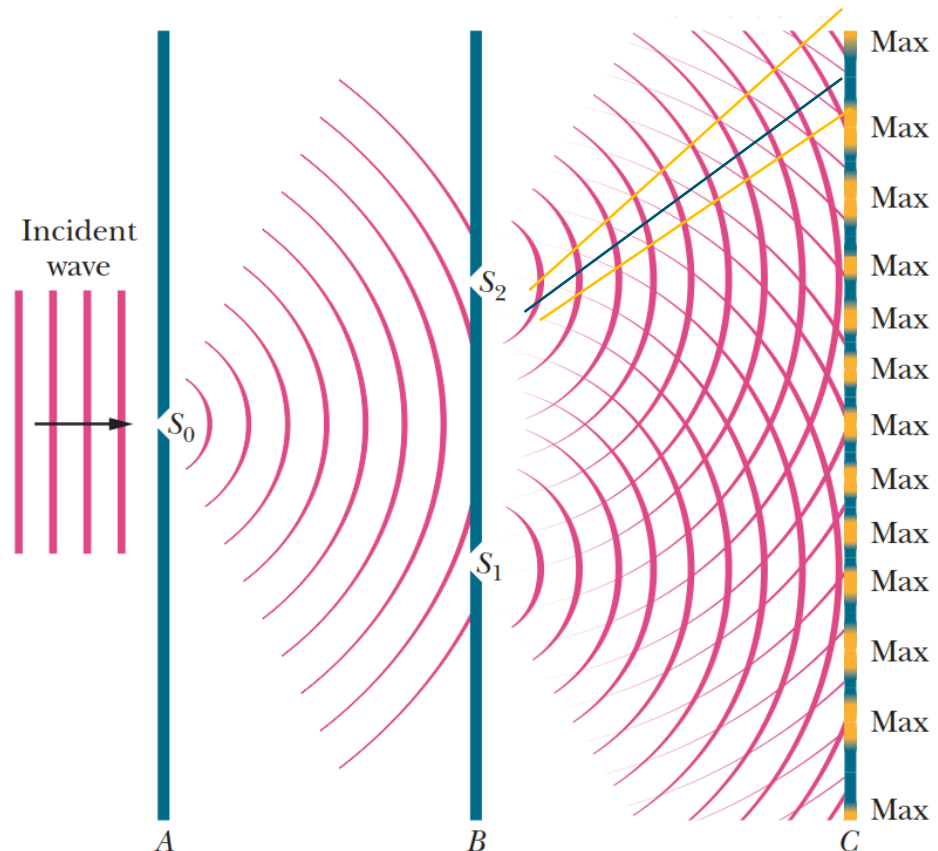
Bright bands (fringes) =

interference maxima;

Dark bands (fringes) =

interference minima.

Bright and dark fringes –
interference pattern.



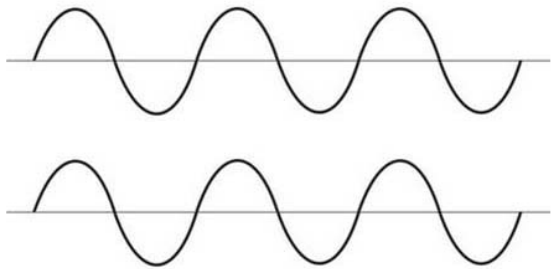
Locating the fringes – approximate geometry

(a) Waves from slits S_1 and S_2 (which extend into and out of the page) combine at P, an arbitrary point on screen C at distance y from the central axis. The angle θ serves as a convenient locator for P.

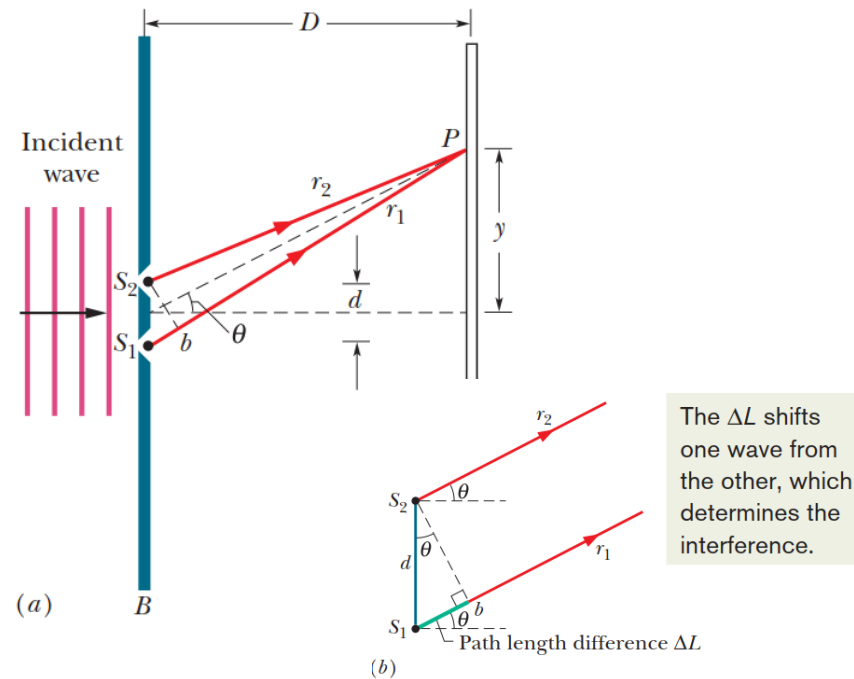
(b) For $D \gg d$, we can approximate rays r_1 and r_2 as being parallel, at angle θ to the central axis

$$\Delta L = d \sin \theta \quad (\text{path length difference})$$

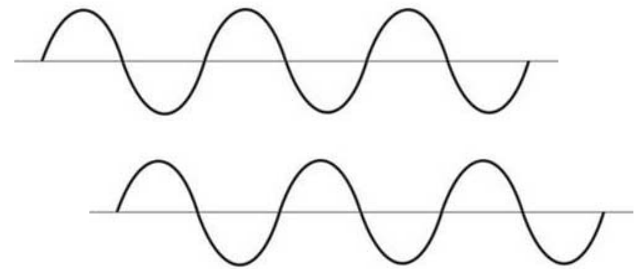
constructive interference



$$d \sin \theta = m \lambda \quad (m = 0, \pm 1, \pm 2, \dots)$$



destructive interference

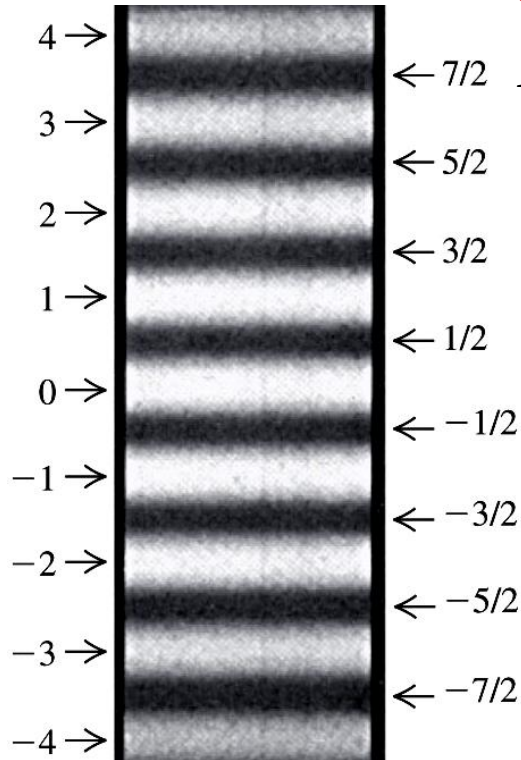


$$d \sin \theta = (m + 0.5) \lambda \quad (m = 0, \pm 1, \pm 2, \dots)$$

Locating the fringes

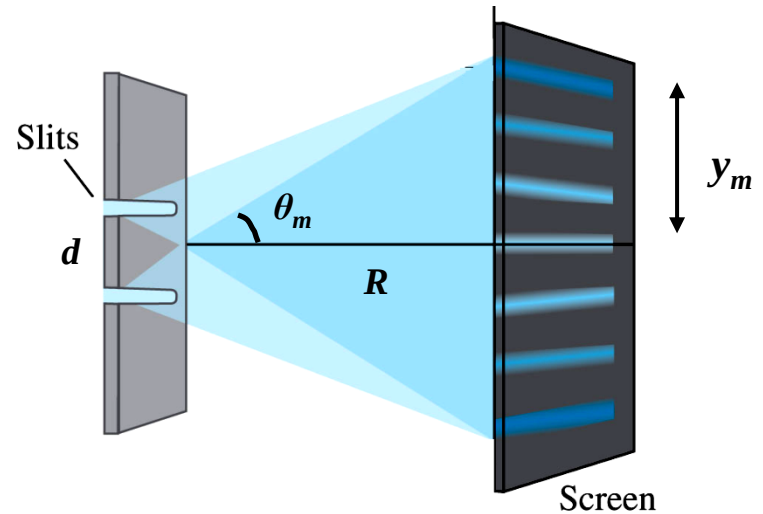
$$d \sin \theta = m \lambda$$

Bright fringes



$$d \sin \theta = (m + 0.5) \lambda$$

Dark fringes



$$y_m = R \tan \theta_m$$

$\tan \theta \approx \sin \theta$
for small θ

$$\rightarrow y_m = R \sin \theta_m$$

$$\text{Minima } y_{\min} = R(m+0.5)\lambda/d$$

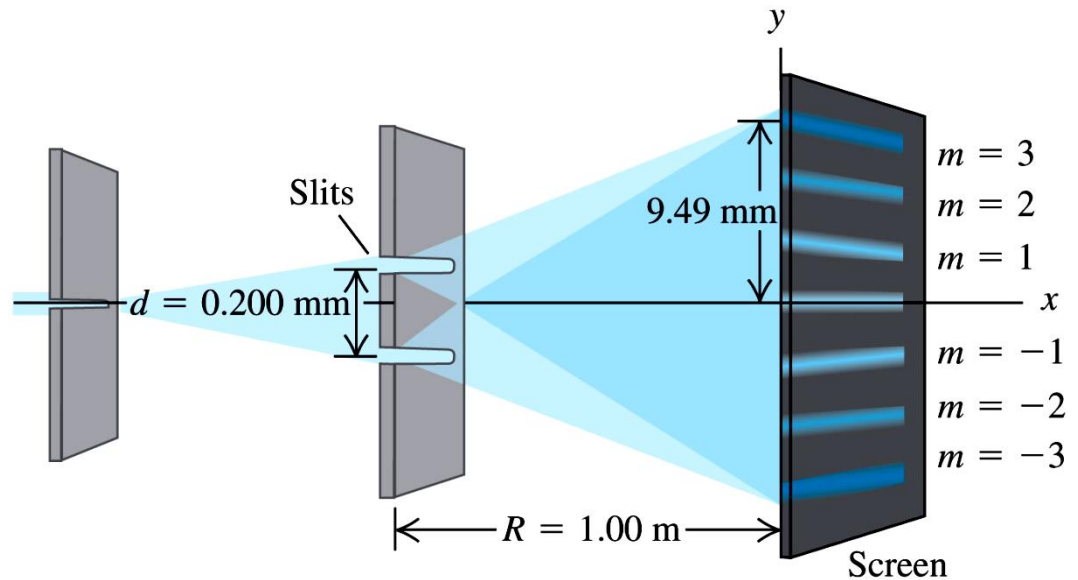
$$\therefore y_m = R \frac{m \lambda}{d} \quad \text{Maxima}$$

In Young's experiment, what is the difference between the distance from S_1 to the $m = +3$ bright area and the distance from S_2 to the same area?



- A. three wavelengths
- B. three half-wavelengths
- C. three quarter-wavelengths
- D. not enough information given to decide

Find the wavelength of a monochromatic light



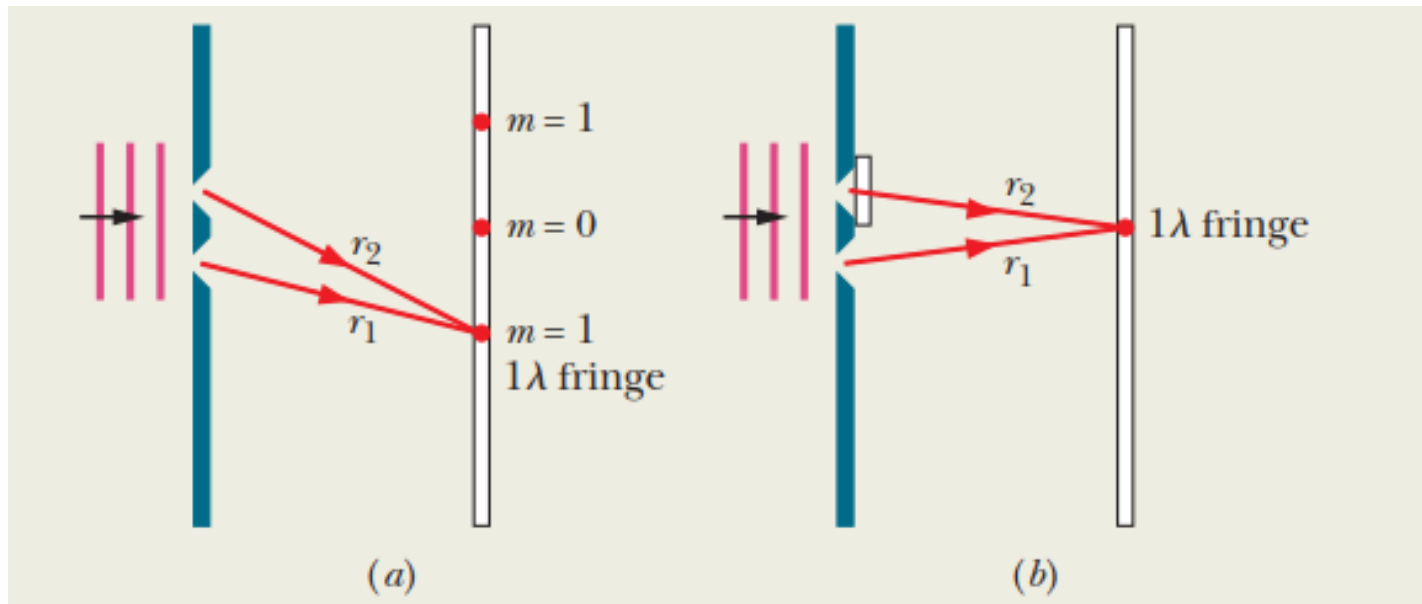
Use $y_m = R \frac{m\lambda}{d}$

$$\begin{aligned}\lambda &= \frac{y_m d}{m R} \\ &= \frac{(9.49 \times 10^{-3} \text{ m})(0.200 \times 10^{-3} \text{ m})}{(3)(1.00 \text{ m})} \\ &= 633 \times 10^{-9} \text{ m} = 633 \text{ nm}\end{aligned}$$

In Young's experiment, if the wavelength of light is increased, how does the pattern change?

-  A. The bright areas move closer together.
- B. The bright areas move farther apart.
- C. The spacing between bright areas remains the same, but the color changes.
- D. any of the above, depending on circumstances
- E. none of the above

A double-slit interference pattern is produced on a screen, as in the figure; the light is monochromatic at a wavelength of 600 nm. A trip of transparent plastic with index of refraction $n = 1.5$ is to be placed over one of the slits. Figure (a) shows the original locations of the $m = 0$ and the $m = 1$ bright fringes. The purpose of the plastic is to shift the pattern upward so that the lower $m = 1$ bright fringe is shifted to the center of the pattern. Should the plastic be placed over the top slit (as arbitrarily drawn in (b)) or the bottom slit, and what thickness L should it have?



光程和光程差

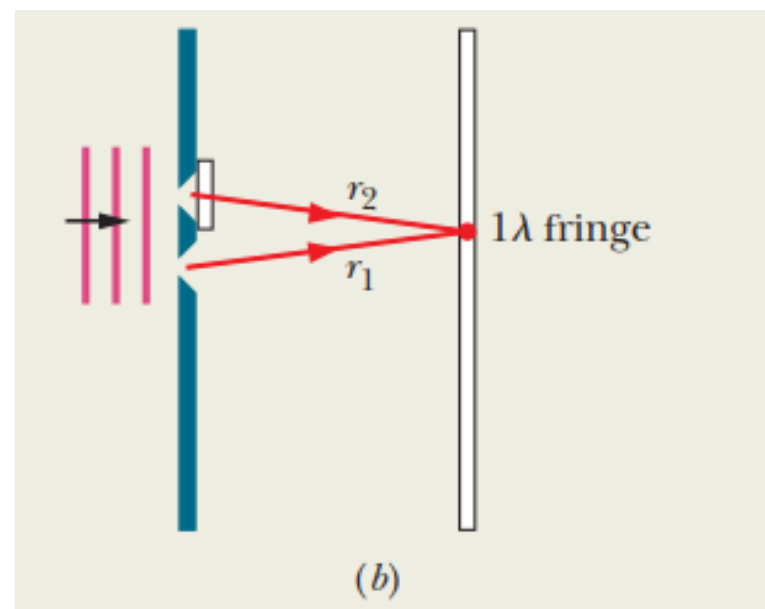
Solution:

Apparently, the path length difference with the plastic is

$$\begin{aligned} \text{OPD} &= \int_{r_2} n(r_2) dr - \int_{r_1} n(r_1) dr \\ &= 1.5 L + 1 \times (r_2 - L) - 1 \times r_1 \\ &= 0.5 L \end{aligned}$$

$$\text{OPD} = 1 \lambda = 0.5 L$$

$$L = \lambda / 0.5 = 600 \text{ nm} / 0.5 = 1.2 \mu\text{m}$$



Interference and double-slit intensity

Here, we limit our case into ***two light waves*** which are

- 1) **Coherent** : monochromatic and with a constant phase relationship
- 2) **With equal amplitude**
- 3) **With same polarization** : the wave disturbances lie along the same plane

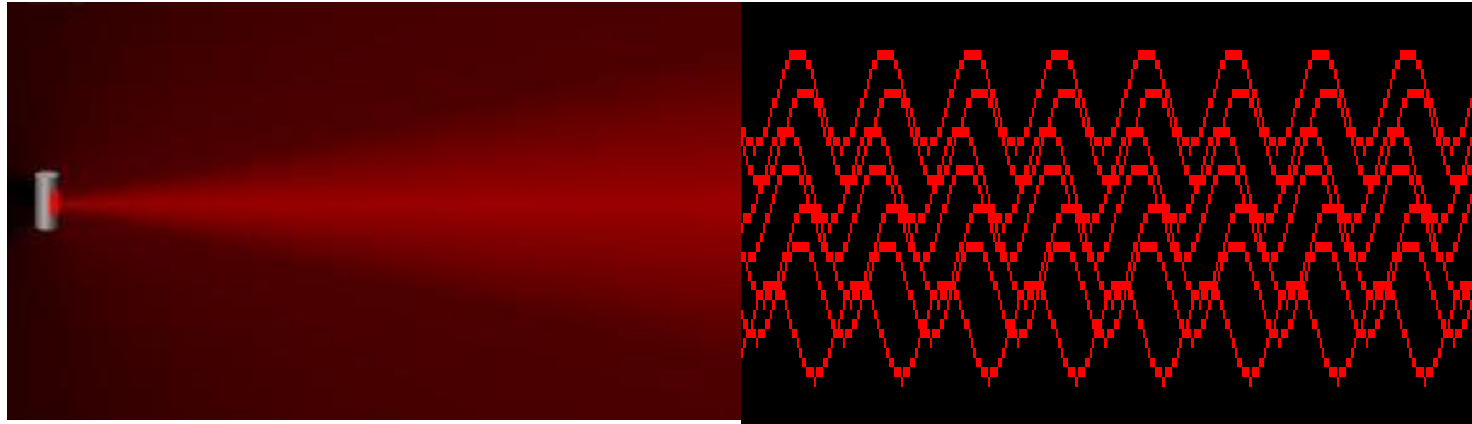


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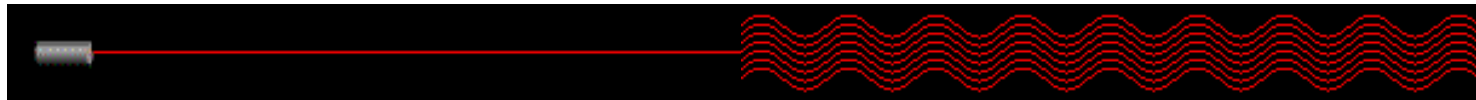
Coherence

When the phase difference between two light source remains constant, they are said to be completely **coherent**. Laser is a coherent light source throughout the area of the source, while light bulbs emit incoherent light. Sunlight is partially coherent.

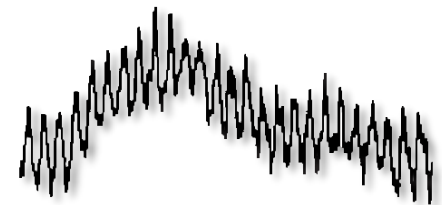
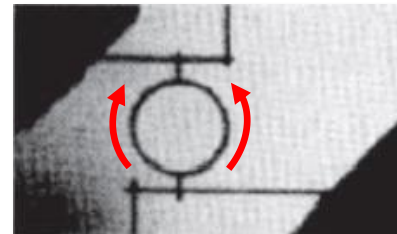
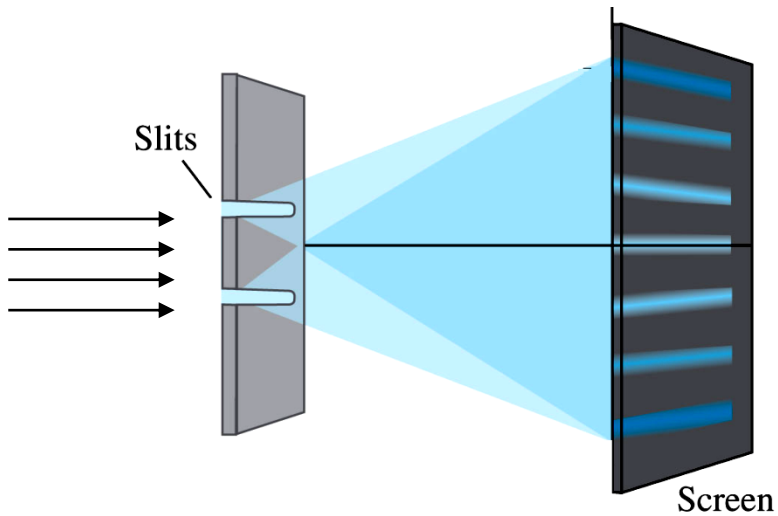
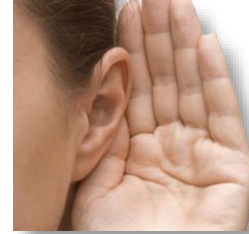
Incoherent source



Coherent source (e.g. laser)

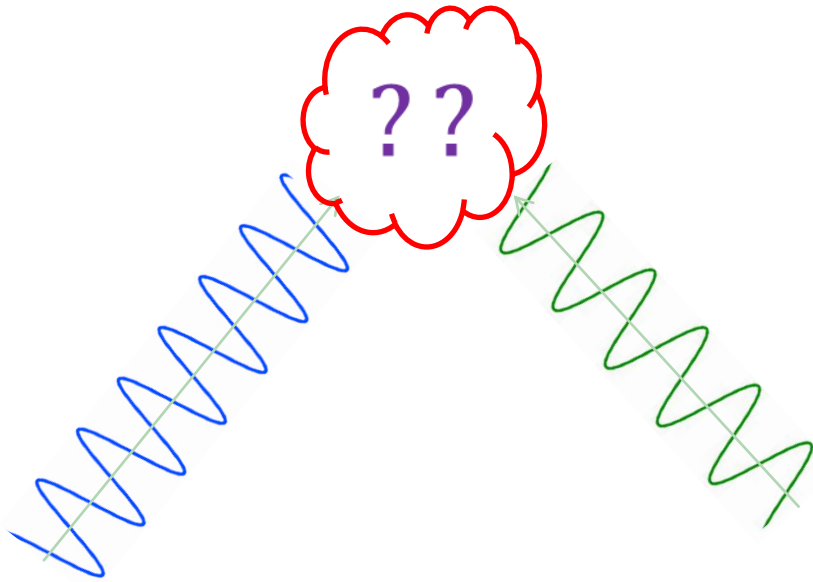


Interference everywhere

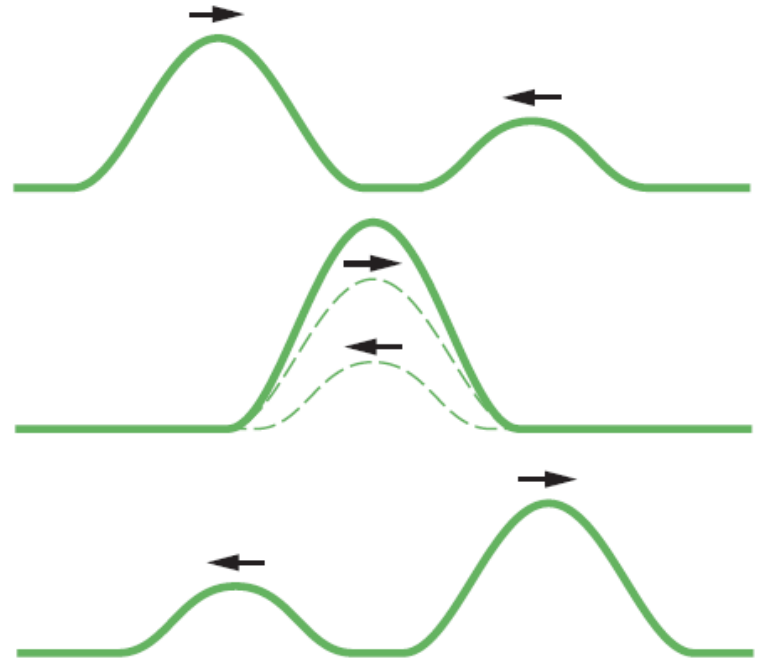


Interference and double-slit intensity

What happens when wave
“meets” wave ?

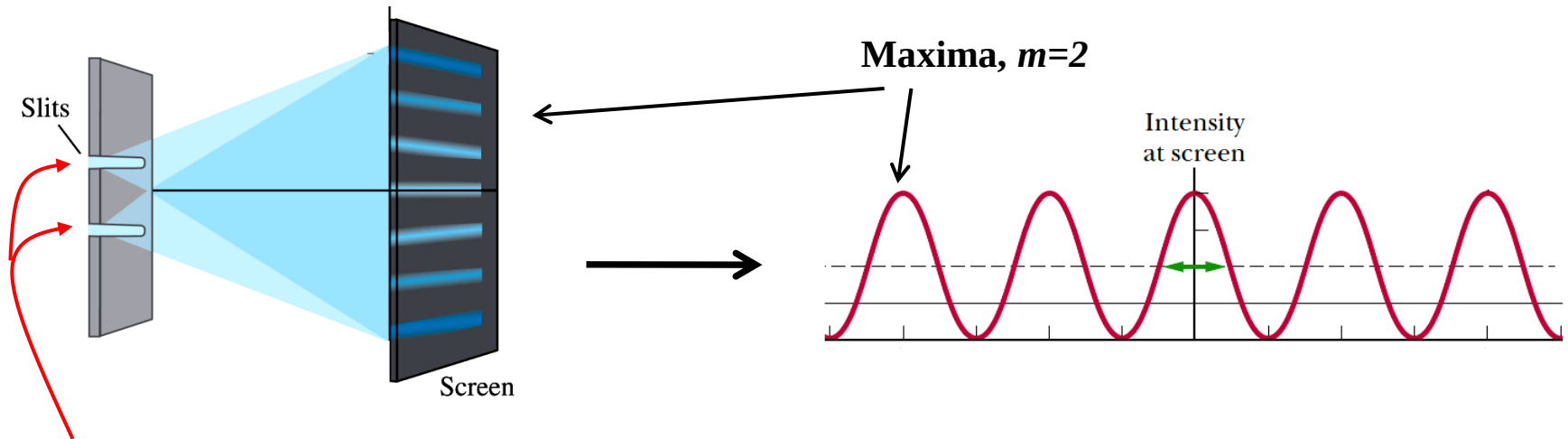


$$y'(x,t) = y_1(x,t) + y_2(x,t)$$



- Overlapping waves algebraically add to produce a resultant wave (net wave).
- Overlapping waves do not in any way alter the travel of each other.

Finding the intensity of the interference pattern

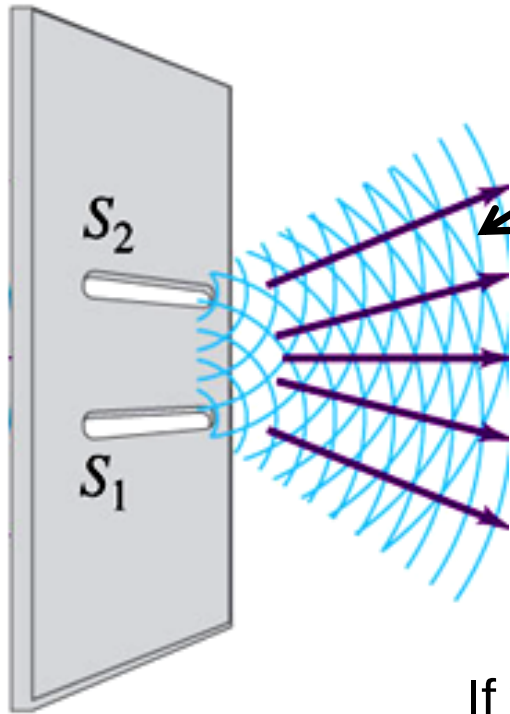


Assuming equal amplitude of E and same polarization

Intensity (power per unit area) of an EM wave : $\epsilon_0 c E^2$

To find the superposition of the two E field, we can use the phasor diagram

Finding the intensity of the interference pattern

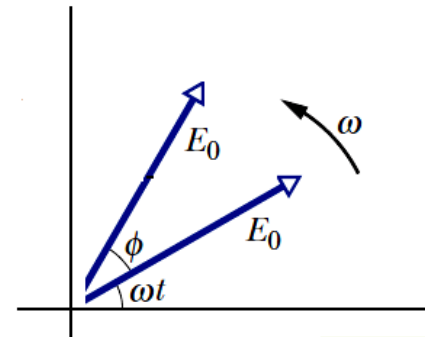


At any given point, there is **a phase difference ϕ** between the E -fields of the two waves, which is a function of position.

$$E_1(t) = E_0 \cos \omega t$$

$$E_2(t) = E_0 \cos(\omega t + \phi)$$

In a phase diagram :



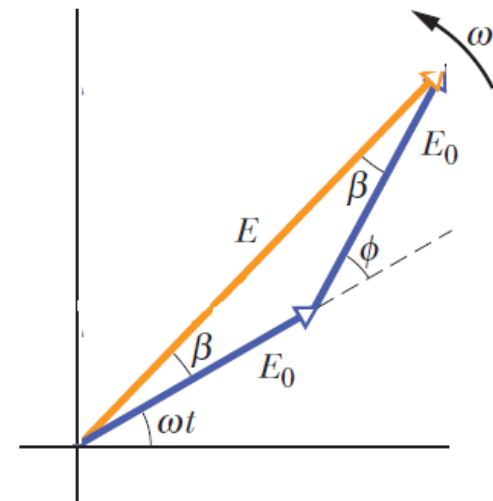
If we add them up by joining them head to tail :

$$\beta + \beta = \phi \quad \rightarrow \quad \beta = \frac{1}{2}\phi$$

Magnitude of E = project of both E_0 on E

$$= 2(E_0 \cos \beta)$$

$$= 2E_0 \cos \frac{1}{2}\phi$$



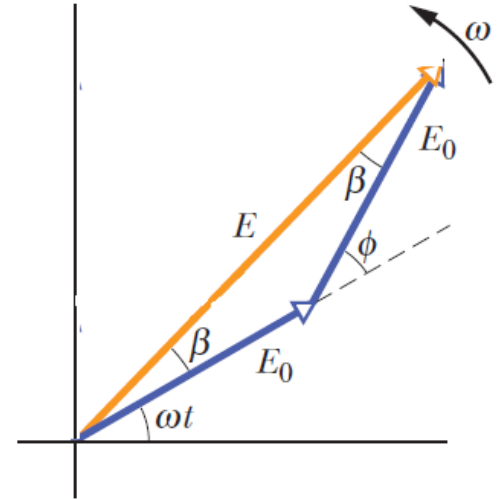
Finding the intensity of the interference pattern

$$\text{Magnitude of } E = 2E_0 \cos \frac{1}{2}\phi$$

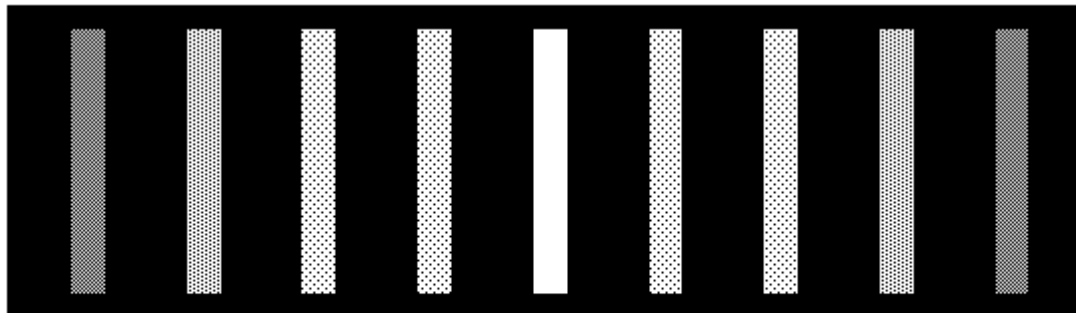
$$E^2 = 4E_0^2 \cos^2 \frac{1}{2}\phi$$

We know $I \propto E^2$ (intensity is proportional to the square of the amplitude)

$$I = 4I_0 \cos^2 \frac{1}{2}\phi$$



Maximum I is 4 times of the original maximum!



Total energy doesn't change, it has just been redistributed. **Prove it!**

A beam of monochromatic light with a wavelength of 660 nm is directed at a double slit. Consider the five fringes labeled in the drawing, including the central maximum that is labeled “B.” Which one of these fringes is produced when the path difference is 1320 nm?

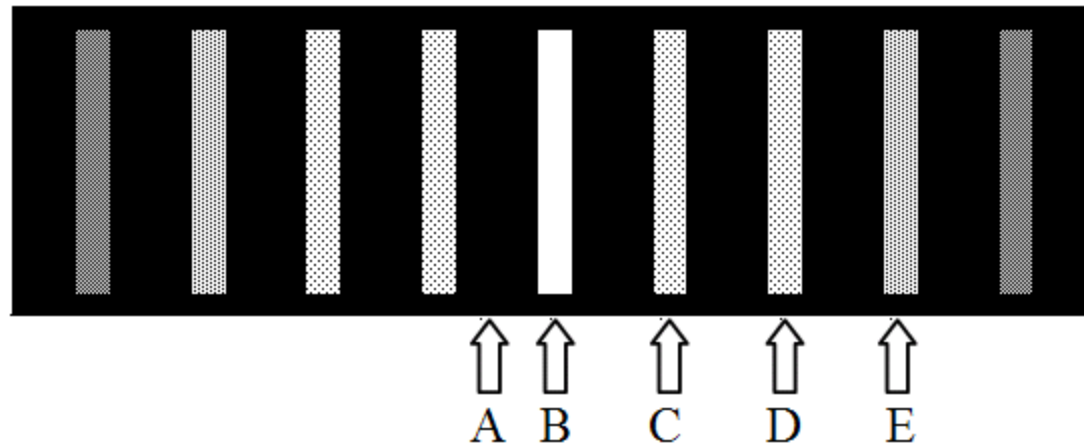
a) A

b) B

c) C

d) D

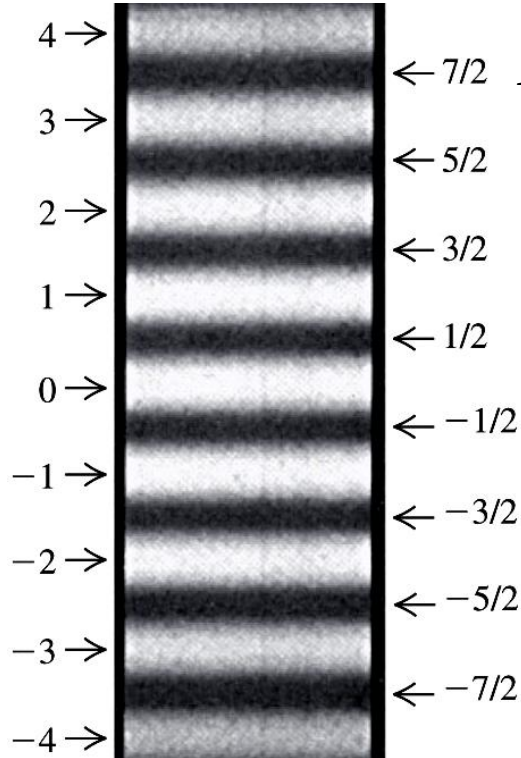
e) E



Locating the fringes

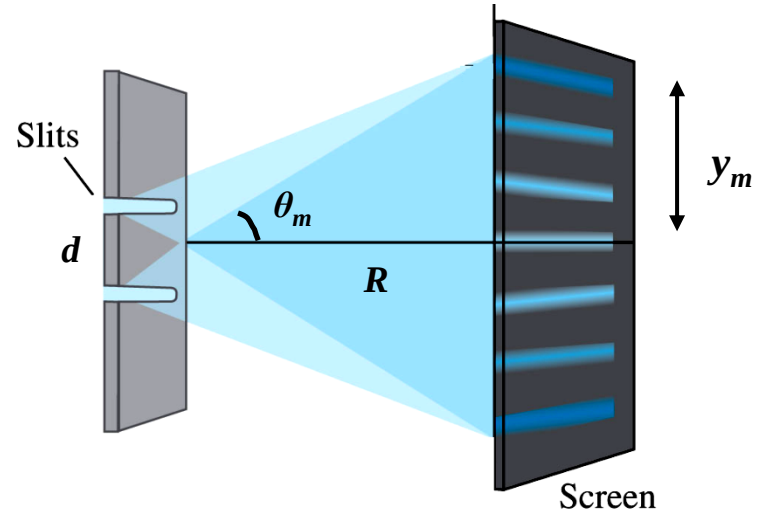
$$d \sin \theta = m \lambda$$

Bright fringes



$$d \sin \theta = (m + 0.5) \lambda$$

Dark fringes



$$y_m = R \tan \theta_m$$

$\tan \theta \approx \sin \theta$
for small θ

$$\rightarrow y_m = R \sin \theta_m$$

$$\therefore y_m = R \frac{m \lambda}{d}$$

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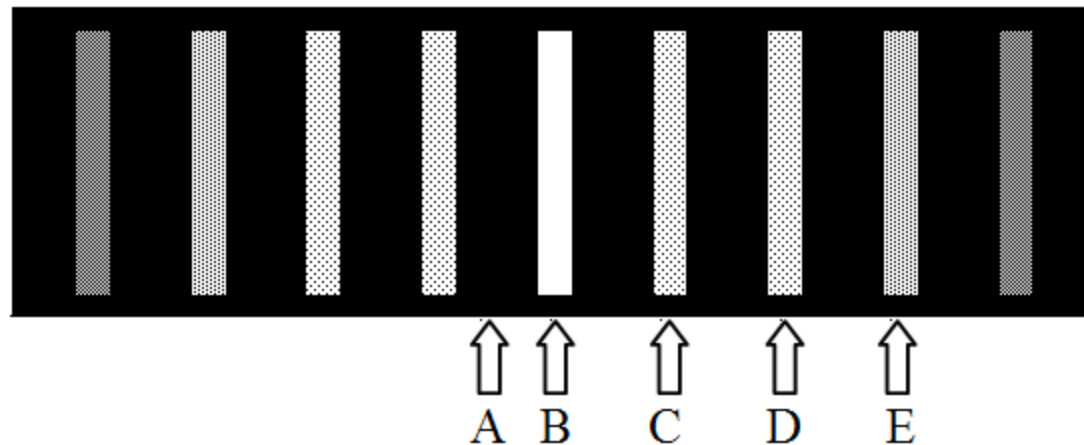
a) A

b) B

c) C

✓ d) D

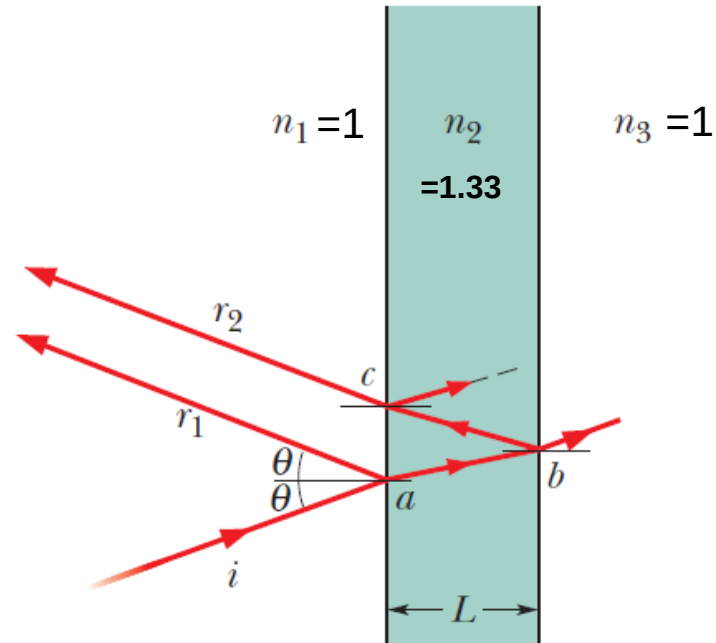
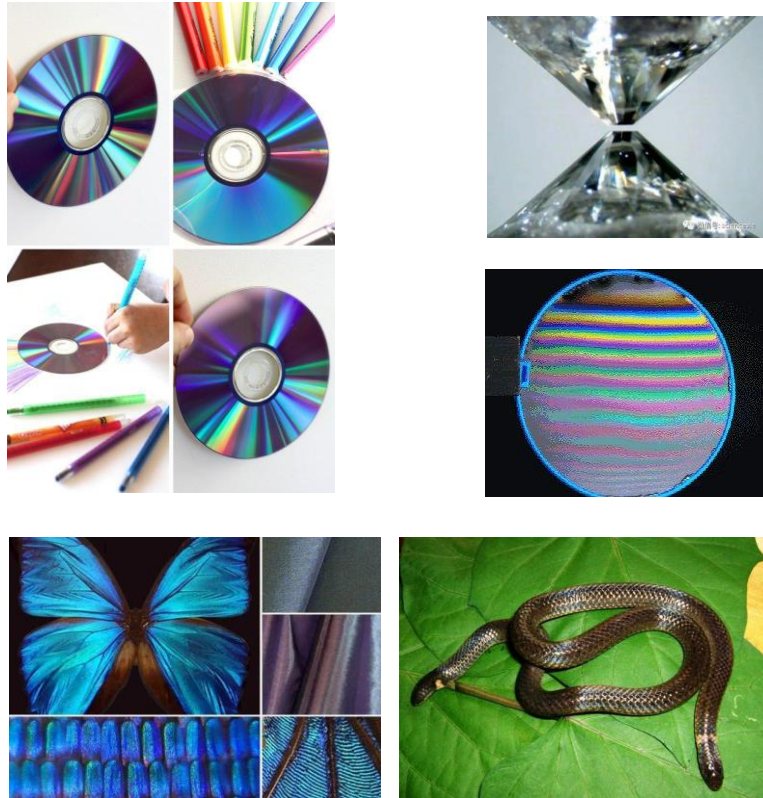
e) E



Today's topics (Chapter 35)

- Interference of Waves
- Double-Slit Experiment
- Thin-Film Interference
- Michelson's Interferometer

Thin-film interference



For simplicity, we assume the incident beam is perpendicular to the surface

$$\text{i.e., } ab + bc \approx 2L$$

~~constructive interference :
 $2L = m\lambda$~~

Two important points:

- 1) wavelength change in medium 2
- 2) There is a phase change at point a!

Reflection of Waves from Boundaries

Reflection from a HARD boundary



At a fixed (hard) boundary, the reflected wave changes its polarity (**undergoes a 180° phase change**)

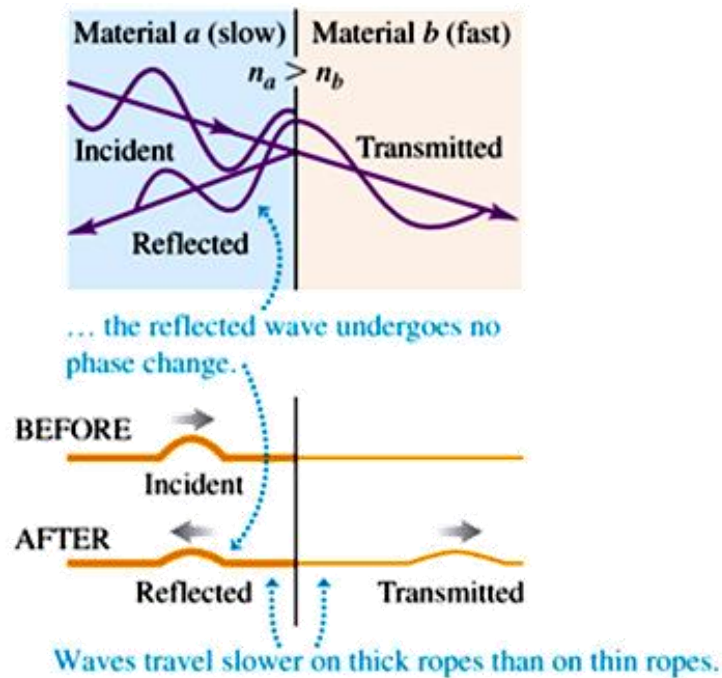
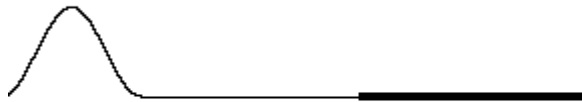
Reflection from a SOFT boundary



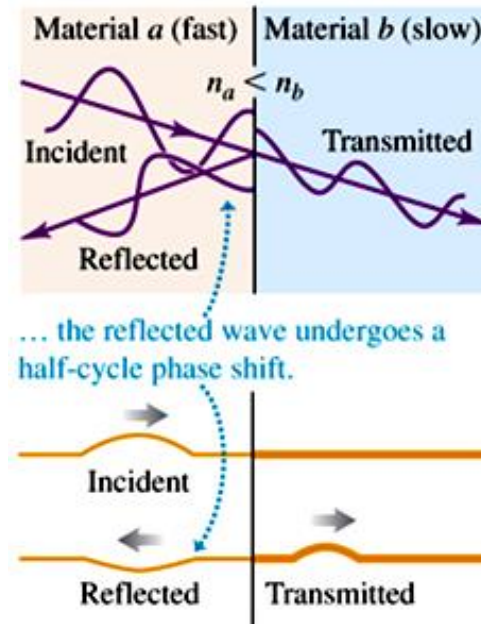
At a free (soft) boundary, the reflected wave has the same polarity (**no phase change**) as the incident wave

Thin-film interference

From high speed to low speed
(low density to high density)



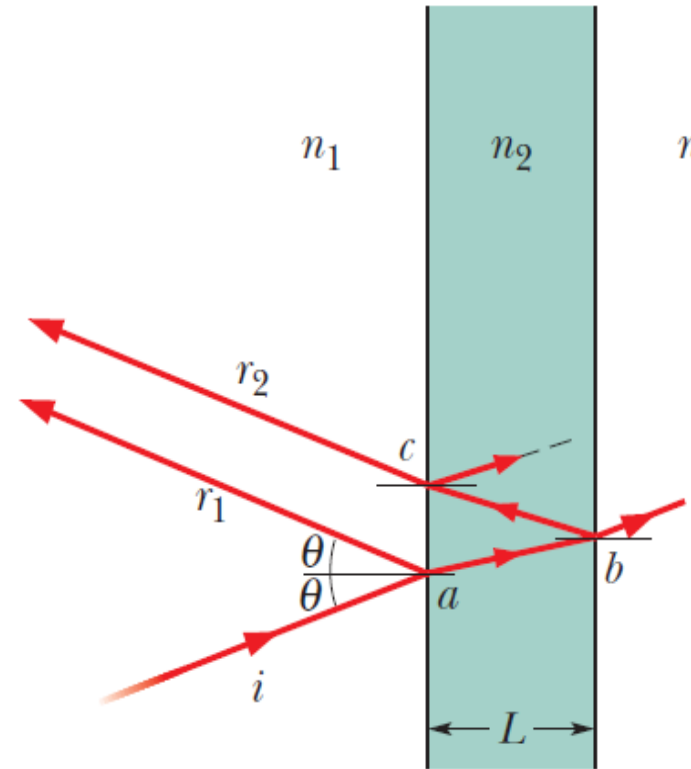
From low speed to high speed
(high density to low density)



Thin-film interference

$$\text{OPD} = n_2(ab+bc) \approx 2n_2L$$

Now, it is critically important to learn that when light (or any waveform) travels from a region with higher velocity (**smaller n**) to a region with lower velocity (**larger n**) and reflects at the interface, the reflected wave will have a phase shift of $\pi/2$. This is known as the half wave loss.



for constructive interference :

$$2L = \left(m + \frac{1}{2}\right) \frac{\lambda}{n_2} \quad \text{for } m = 0, 1, 2, \dots$$

for destructive interference :

$$2L = m \frac{\lambda}{n_2} \quad \text{for } m = 0, 1, 2, \dots$$

Thin-film interference



if $L \ll \lambda$

we can ignore the thickness of the film

Always destructive interference as the phase difference is always π

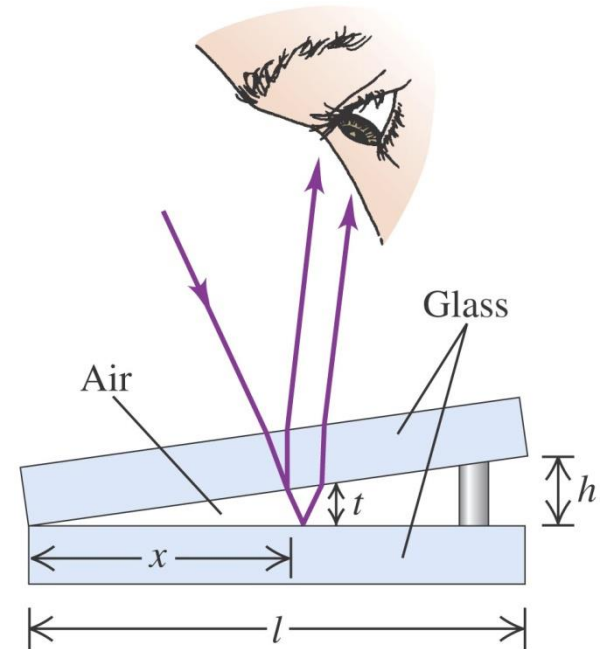
Table 35-1 An Organizing Table for Thin-Film Interference in Air (Fig. 35-17)^a

	r_1	r_2
Reflection phase shifts	0.5 wavelength	0
Path length difference	$2L$	
Index in which path length difference occurs	n_2	
In phase ^a :	$2L = \frac{\text{odd number}}{2} \times \frac{\lambda}{n_2}$	
Out of phase ^a :	$2L = \text{integer} \times \frac{\lambda}{n_2}$	

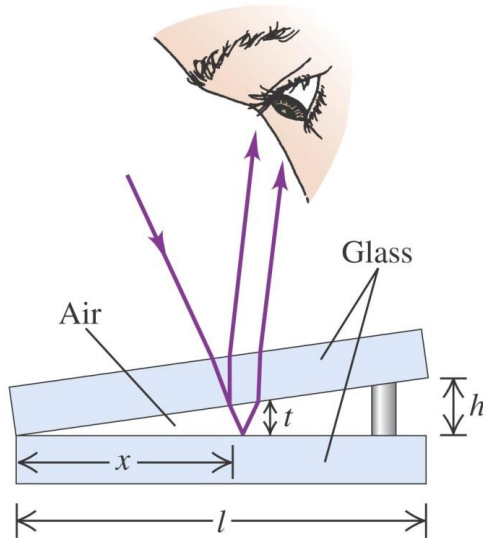
^aValid for $n_2 > n_1$ and $n_2 > n_3$.

An air wedge separates two glass plates as shown. Light of wavelength λ strikes the upper plate at normal incidence. At a point where the air wedge has thickness t , you will see a bright fringe if t equals

- ✓ A. $\lambda/2$.
- B. $3\lambda/4$.
- C. λ .
- D. either A. or C.
- E. any of A., B., or C.



Thin film vs. thick film

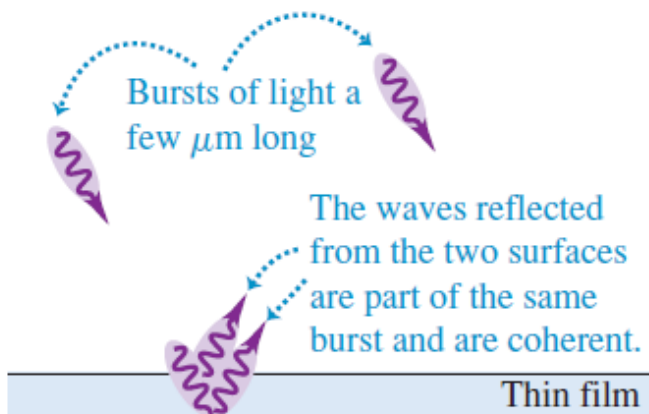


Can you see the fringes if h is too large? Why?

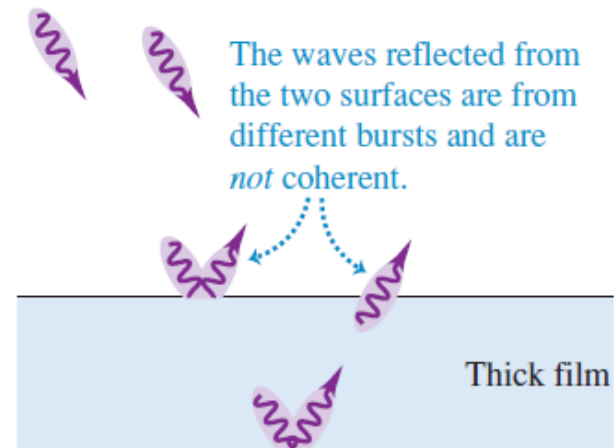
Nope! For two reasons!

1. Fringes become too dense.
2. Light may not be coherent any more for large h .

(a) Light reflecting from a thin film

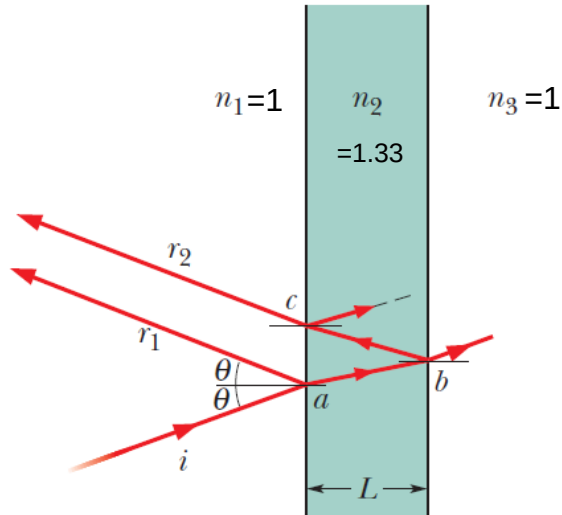


(b) Light reflecting from a thick film



Application of thin film interference

Thin film interference is often used to measure thickness of thin films.



constructive interference:

$$2L = (m + \frac{1}{2}) \frac{\lambda}{n_2} \quad \text{for } m = 0, 1, 2, \dots$$

destructive interference:

$$2L = m \frac{\lambda}{n_2} \quad \text{for } m = 0, 1, 2, \dots$$

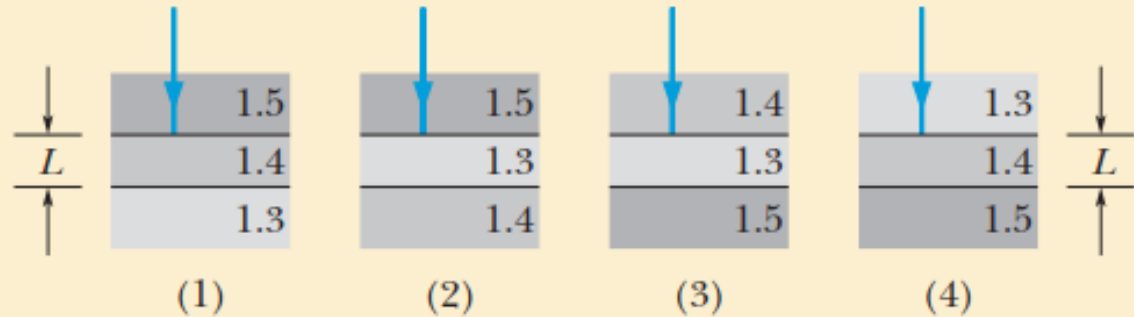
Matter waves (such as electron waves) can be also used for such application!

But what the heck is matter wave? To be continued.....



Checkpoint 5

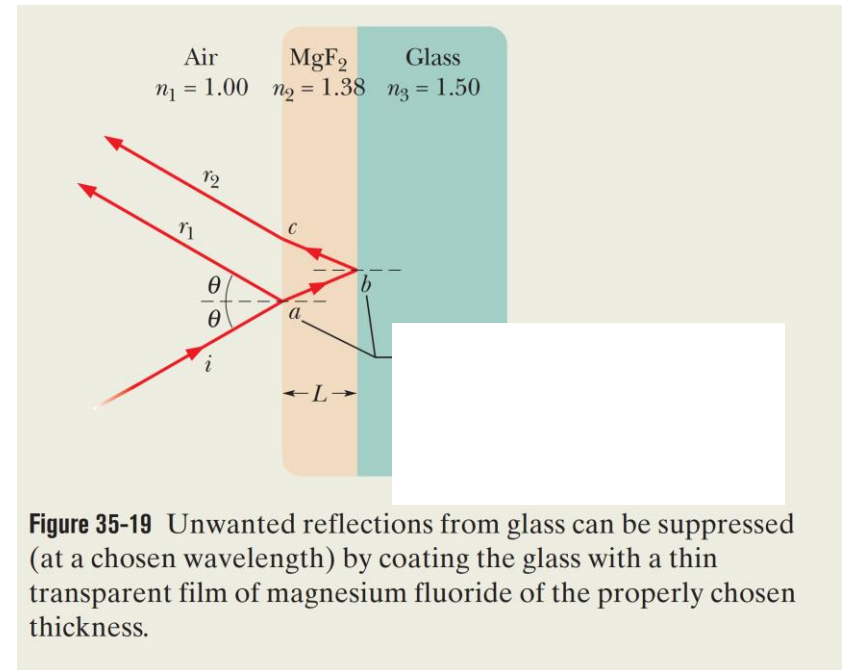
The figure shows four situations in which light reflects perpendicularly from a thin film of thickness L ,



with indexes of refraction as given. (a) For which situations does reflection at the film interfaces cause a zero phase difference for the two reflected rays? (b) For which situations will the film be dark if the path length difference $2L$ causes a phase difference of 0.5 wavelength?

Always remember that when n goes from small to big, there's a half wave loss (HWL).

A glass lens is coated on one side with a thin film of magnesium fluoride (MgF_2) to reduce reflection from the lens surface. The index of refraction of MgF_2 is 1.38; that of the glass is 1.50. What is the least coating thickness that eliminates (via interference) the reflections at the middle of the visible spectrum (550 nm)? Assume that the light is approximately perpendicular to the lens surface.



$$\text{OPD} = 2n_2L \quad \frac{\text{OPD}}{\lambda} = \frac{1}{2}, \frac{3}{2}, \dots \Rightarrow L = \left(\frac{1}{2}, \frac{3}{2}, \dots \right) \times \frac{\lambda}{2n_2}$$

$$L_{\min} = \frac{1}{2} \frac{\lambda}{2n_2} = \frac{550 \text{ nm}}{4 \times 1.38} = 99.6 \text{ nm}.$$

Always remember that when n goes from small to big, there's a half wave loss (HWL).

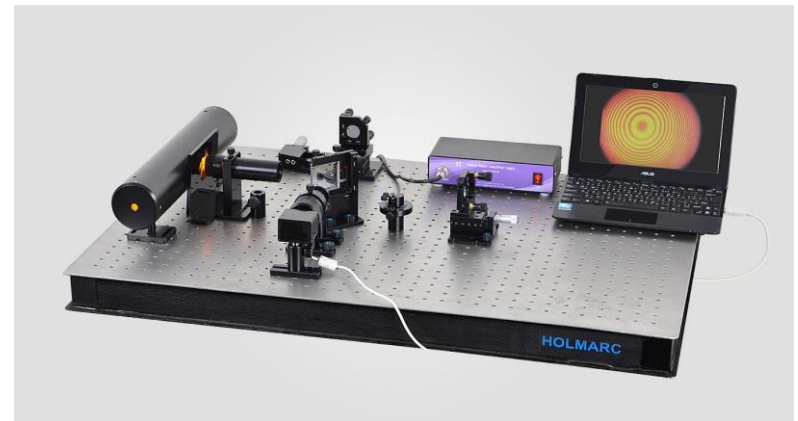
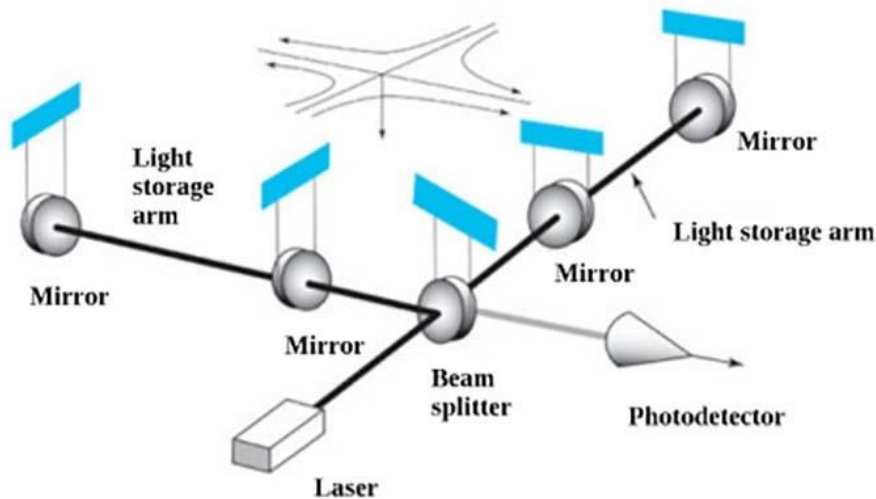
Today's topics (Chapter 35)

- Interference of Waves
- Double-Slit Experiment
- Thin-Film Interference
- Michelson's Interferometer

Michelson's interferometer

The **Michelson interferometer** is a common configuration for optical interferometry. Using a beamsplitter, a light source is split into two arms. Each of those is reflected back toward the beamsplitter which then combines their amplitudes interferometrically. The resulting interference pattern that is not directed back toward the source is typically directed to some type of photoelectric detector or camera.

Basic interferometer design



Michelson's interferometer

Michelson interferometer

The path length difference for the two waves when they recombine at the telescope is $2d_2 - 2d_1$, and anything that changes this path length difference (**OPD**) will cause a change in the phase difference between these two waves at the eye. Ways to change the OPD are:

1. Change d_2 :

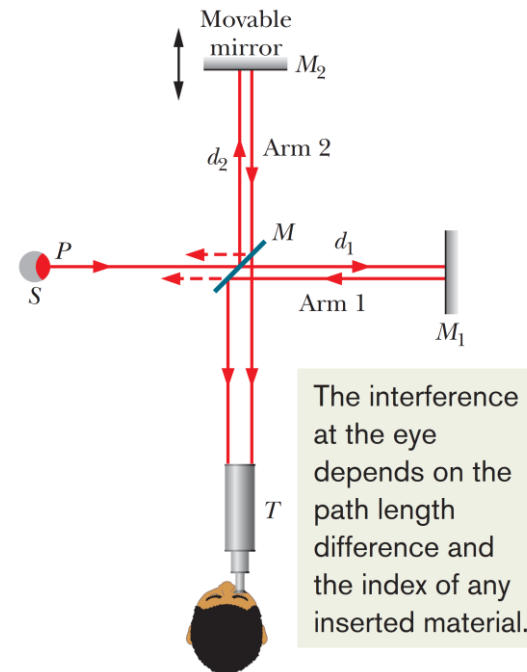
$$\Delta \text{OPD} = 2\Delta d_2$$

2. Insert a transparent medium (refraction index n , width L) in Arm 1 or Arm 2:

$$\Delta \text{OPD} = 2nL$$



Albert Michelson (1852 –1931) , the first American to win Nobel Prize in a science! (in 1907)



Michelson-Morley experiment

From Chapter 33 we derived from Maxwell's equations that

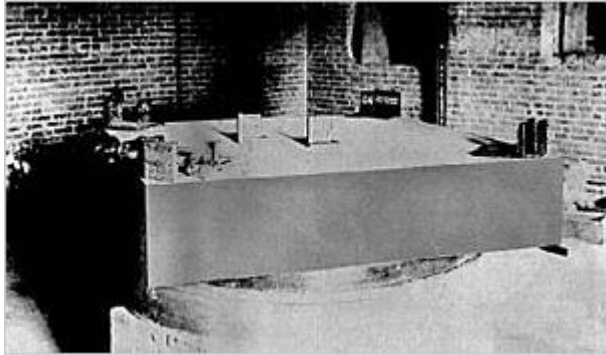
$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 299\,792\,458 \text{ m/s.}$$

If a reference frame (rf) has velocity v on another rf, will c become $c + v$? If so, Maxwell's equations will work at only one frame, called the “aether” frame.

The **Michelson-Morley experiment** used the Michelson interferometer to measure the speed of light in perpendicular directions, in an attempt to detect the relative motion of matter through the stationary “aether”. It proved that the interference pattern between the two arms does NOT change no matter how they are oriented. The earth is NOT moving against some frame. All internal reference frames are equal.

For the glory of aether!

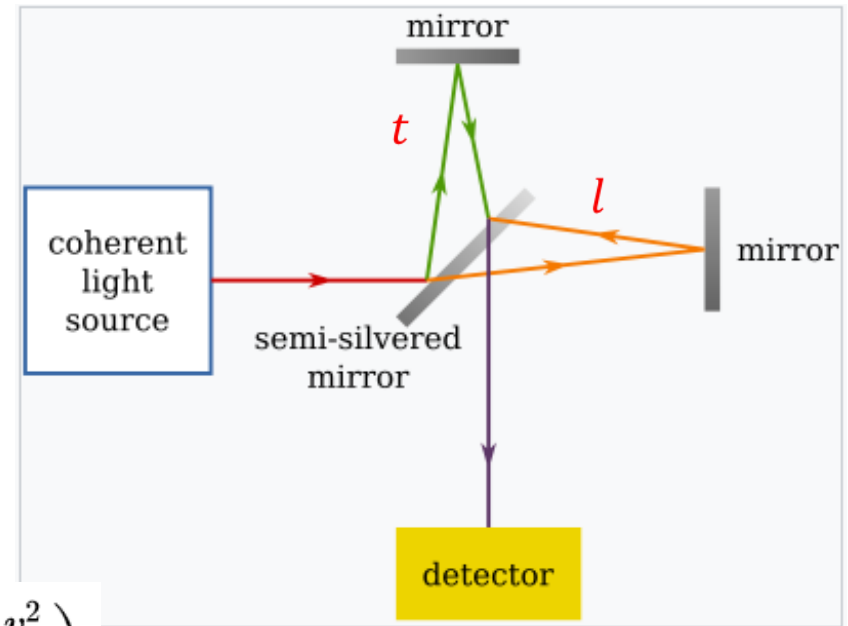
Michelson–Morley experiment (1887), the most famous “failed” experiment



Earth orbits around the Sun at 30 km/s. So aether wind $v = 30$ km/s

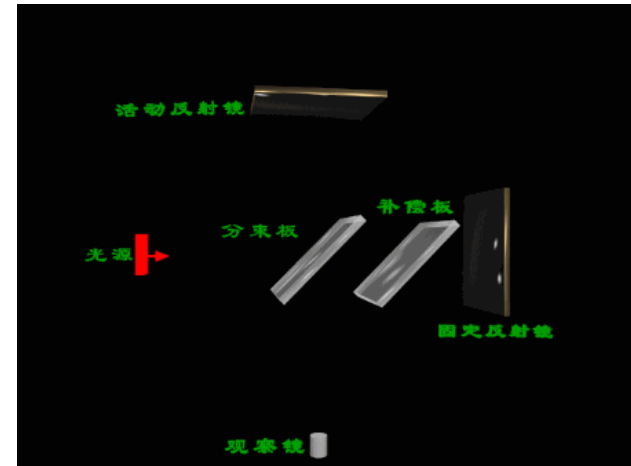
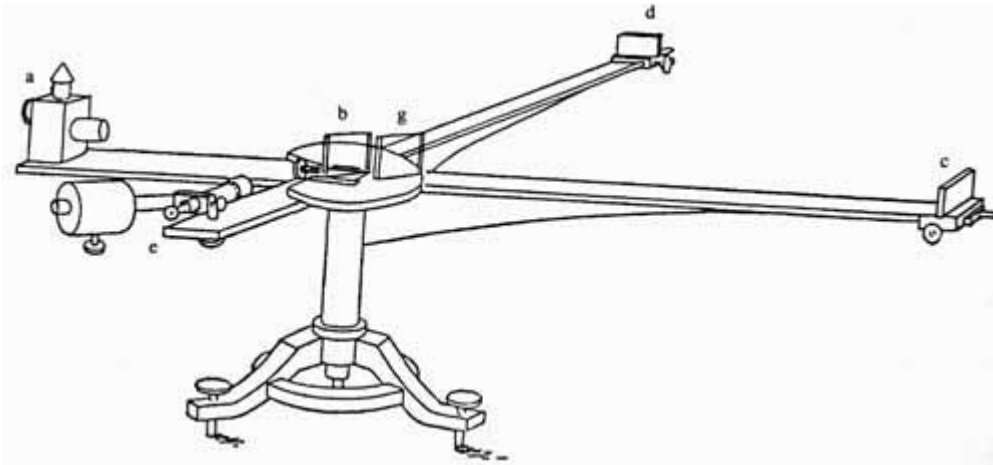
$$T_l = \frac{L}{c-v} + \frac{L}{c+v} = \frac{2L}{c} \frac{1}{1 - \frac{v^2}{c^2}} \approx \frac{2L}{c} \left(1 + \frac{v^2}{c^2} \right)$$
$$T_t = \frac{2L}{\sqrt{c^2 - v^2}} = \frac{2L}{c} \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \approx \frac{2L}{c} \left(1 + \frac{v^2}{2c^2} \right)$$

We all know what happened next...



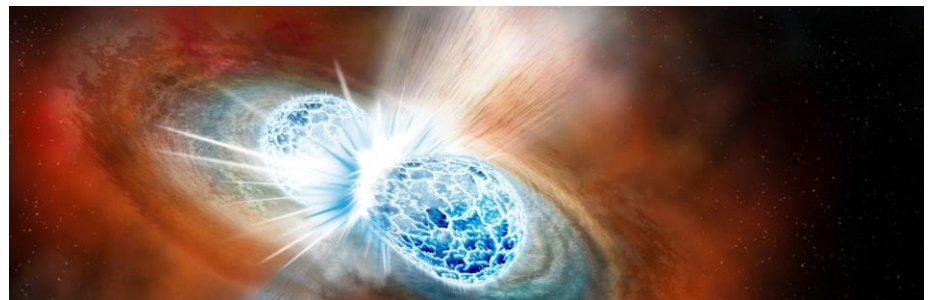
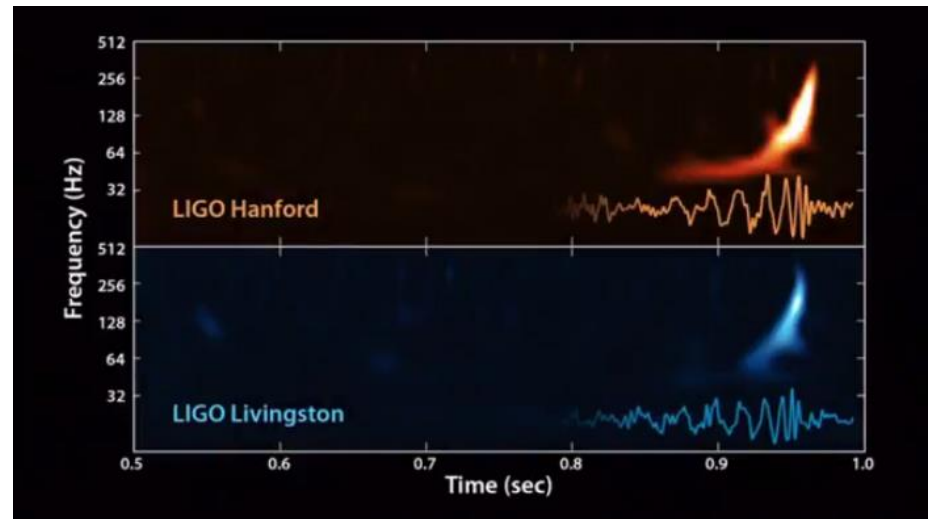
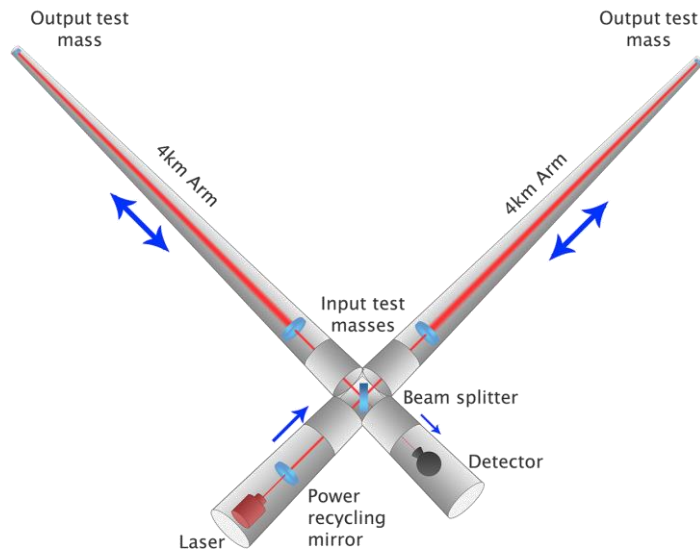
Michelson-Morley experiment

The negative result of the Michelson-Morley experiment is generally considered to be the first strong evidence against the then-prevalent aether theory, and indicated a line of research that eventually led to special relativity, which rules out a stationary aether. The experiment has been referred to as “the moving-off point for the theoretical aspects of the Second Scientific Revolution”



Michelson-Morley experiment: Detection of the gravitational wave

The first observation of gravitational waves was also done with a Michelson-type interferometer. Announced by the LIGO and Virgo collaborations on 11 Feb. 2016, a gravitational wave emanating from the inward spiral and merger of a pair of black holes was detected



Huygens' Principle The three-dimensional transmission of waves, including light, may often be predicted by *Huygens' principle*, which states that all points on a wavefront serve as point sources of spherical secondary wavelets. After a time t , the new position of the wavefront will be that of a surface tangent to these secondary wavelets.

The law of refraction can be derived from Huygens' principle by assuming that the index of refraction of any medium is $n = c/v$, in which v is the speed of light in the medium and c is the speed of light in vacuum.

Wavelength and Index of Refraction The wavelength λ_n of light in a medium depends on the index of refraction n of the medium:

$$\lambda_n = \frac{\lambda}{n}, \quad (35-6)$$

in which λ is the wavelength in vacuum. Because of this dependency, the phase difference between two waves can change if they pass through different materials with different indexes of refraction.

Young's Experiment In *Young's interference experiment*, light passing through a single slit falls on two slits in a screen. The light leaving these slits flares out (by diffraction), and interference occurs in the region beyond the screen. A fringe pattern, due to the interference, forms on a viewing screen.

The light intensity at any point on the viewing screen depends in part on the difference in the path lengths from the slits to that point. If this difference is an integer number of wavelengths, the waves interfere constructively and an intensity maximum results. If it is an odd number of half-wavelengths, there is destructive interference and an intensity minimum occurs. The conditions for maximum and minimum intensity are

$$d \sin \theta = m\lambda, \quad \text{for } m = 0, 1, 2, \dots \quad (\text{maxima—bright fringes}), \quad (35-14)$$

$$d \sin \theta = (m + \frac{1}{2})\lambda, \quad \text{for } m = 0, 1, 2, \dots \quad (\text{minima—dark fringes}), \quad (35-16)$$

where θ is the angle the light path makes with a central axis and d is the slit separation.

Coherence If two light waves that meet at a point are to interfere perceptibly, the phase difference between them must remain constant with time; that is, the waves must be **coherent**. When two coherent waves meet, the resulting intensity may be found by using phasors.

Intensity in Two-Slit Interference In Young's interference experiment, two waves, each with intensity I_0 , yield a resultant wave of intensity I at the viewing screen, with

$$I = 4I_0 \cos^2 \frac{1}{2} \phi, \quad \text{where } \phi = \frac{2\pi d}{\lambda} \sin \theta. \quad (35-22, 35-23)$$

Equations 35-14 and 35-16, which identify the positions of the fringe maxima and minima, are contained within this relation.

Thin-Film Interference When light is incident on a thin transparent film, the light waves reflected from the front and back surfaces interfere. For near-normal incidence, the wavelength conditions for maximum and minimum intensity of the light reflected from a *film in air* are

$$2L = (m + \frac{1}{2}) \frac{\lambda}{n_2}, \quad \text{for } m = 0, 1, 2, \dots \quad (\text{maxima—bright film in air}), \quad (35-36)$$

$$2L = m \frac{\lambda}{n_2}, \quad \text{for } m = 0, 1, 2, \dots \quad (\text{minima—dark film in air}), \quad (35-37)$$

where n_2 is the index of refraction of the film, L is its thickness, and λ is the wavelength of the light in air.

If the light incident at an interface between media with different indexes of refraction is in the medium with the smaller index of refraction, the reflection causes a phase change of π rad, or half a wavelength, in the reflected wave. Otherwise, there is no phase change due to the reflection. Refraction causes no phase shift.

The Michelson Interferometer In *Michelson's interferometer* a light wave is split into two beams that, after traversing paths of different lengths, are recombined so they interfere and form a fringe pattern. Varying the path length of one of the beams allows distances to be accurately expressed in terms of wavelengths of light, by counting the number of fringes through which the pattern shifts because of the change.